

Basic Terms and Concepts of Mechanical Ventilation

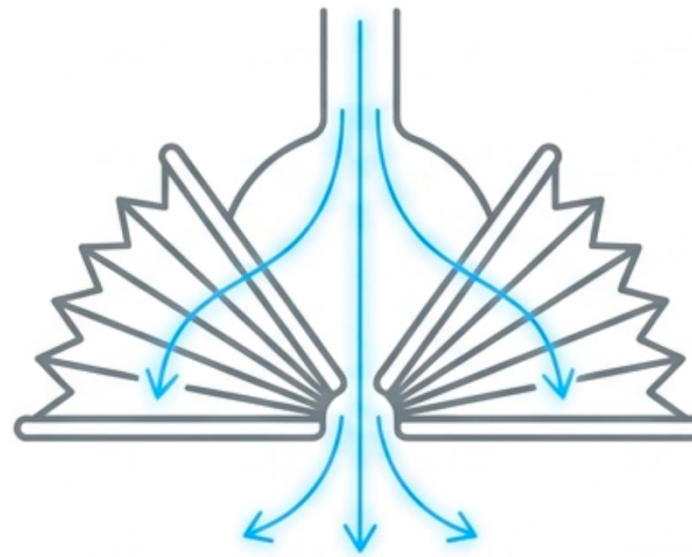
Core Course Concepts

- Physiology of Natural Breathing
- Pressure & Pressure Gradients in the Lungs
- Types of Mechanical Ventilation (PPV, NPV, HFV)
- Lung Mechanics: Resistance, Compliance and Time Constance

Readings

1. Pilbeam's: Chapter 1
2. Egan's: Chapters 47

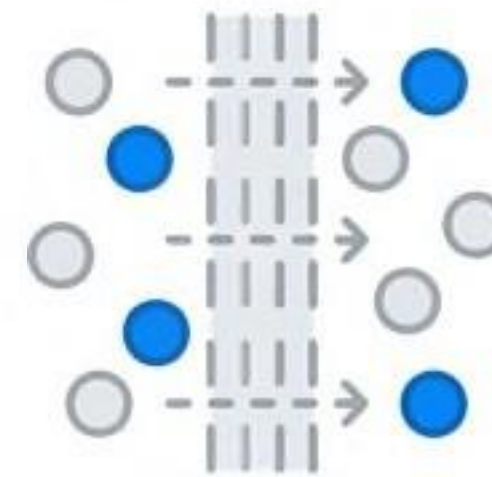
Ventilation vs. Respiration:



VENTILATION

The mechanical pump. Involves the **Bulk flow** of air into and out of the lungs (spontaneous or mechanical)

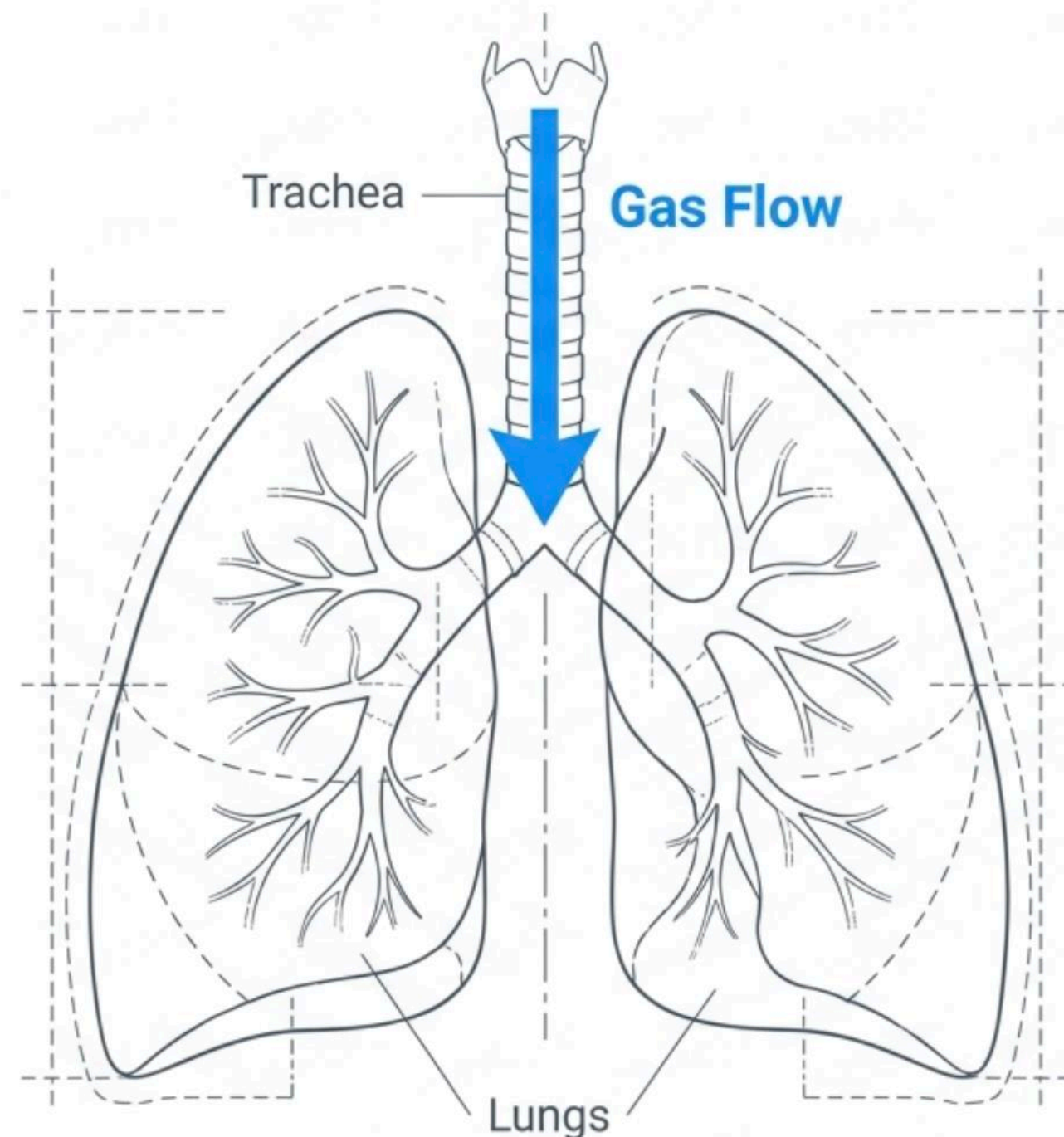
自主



RESPIRATION

Molecular chemical gas exchange crossing the Alveolar-Capillary (A-C) membrane.

The Driving Forces of Ventilation

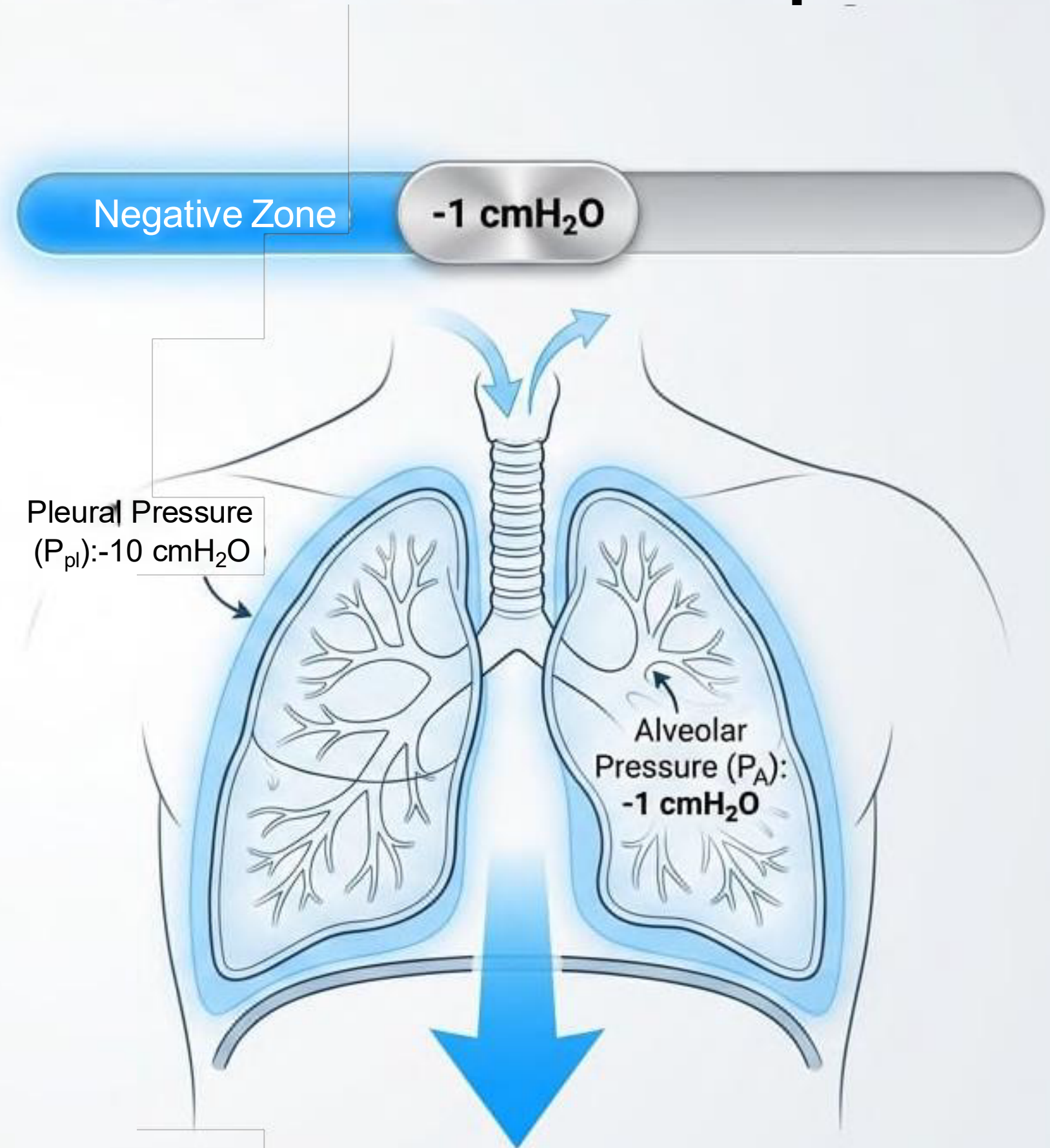


- Gas flow is fundamentally driven by **pressure gradients** — air always flows from an area of higher pressure to an area of lower pressure. 壓差
- When the pressure at the airway opening and the alveoli are **perfectly equal**, **bulk gas flow stops**.
- This **equilibrium** 平衡 (**zero flow**) naturally occurs at two distinct points in the unassisted respiratory cycle:
 - End of inspiration
 - End of exhalation兩邊壓差相等時沒有流量

Inspiration: How Alveolar Pressure Becomes Subatmospheric

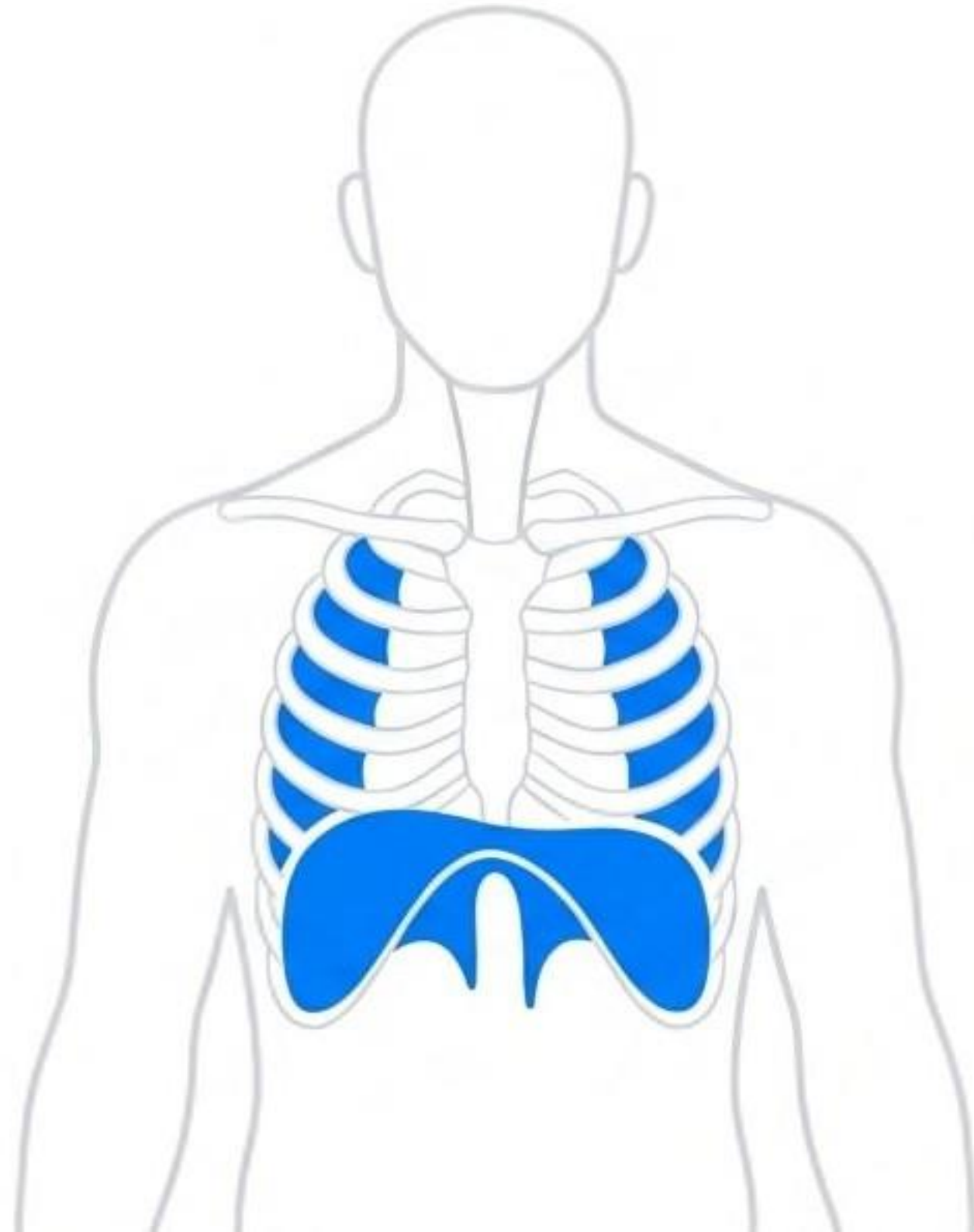
- Thorax expands, causing pleural pressure (P_{pl}) to drop from -5 to -10 cmH₂O.
- Alveolar pressure (P_A) becomes subatmospheric (-1 cmH₂O), pulling air in.

心膈、外肋間肌收縮



Primary Muscles of Spontaneous Inspiration

吸氣屬於主動運動



① Diaphragm 上下拉開

Innervated by the phrenic nerve; descends to expand the thorax vertically.

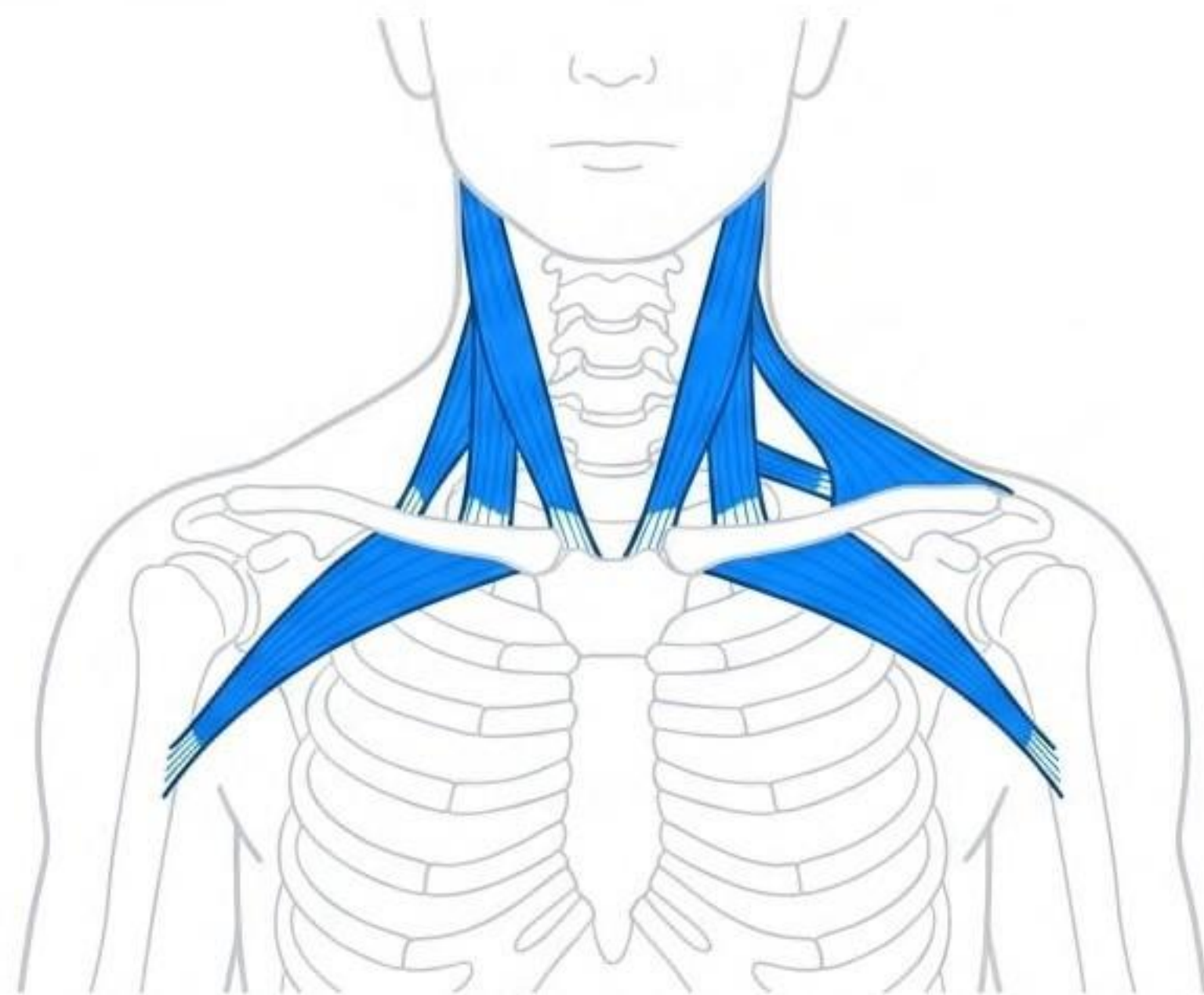
② 外肋間肌 External Intercostals 左右拉開

Elevate the ribs to increase the circumference of the thorax.

Core Function:

Provides the **active** energy to overcome the elastic and resistive loads of the respiratory system.

Accessory Muscles of Inspiration: Emergency Support



Emergency Muscle Group

Scalenes (anterior, medial, posterior) 斜角

Sternocleidomastoids 胸鎖乳突肌

Pectoralis (major and minor) Trapezius 胸小

Clinical Insight

Recruitment strictly indicates a significantly increased Work of Breathing (WOB). The body is fighting severe impedance.

Exhalation: Passive Recoil vs. Active Compression

吐氣屬於被動

Quiet Breathing

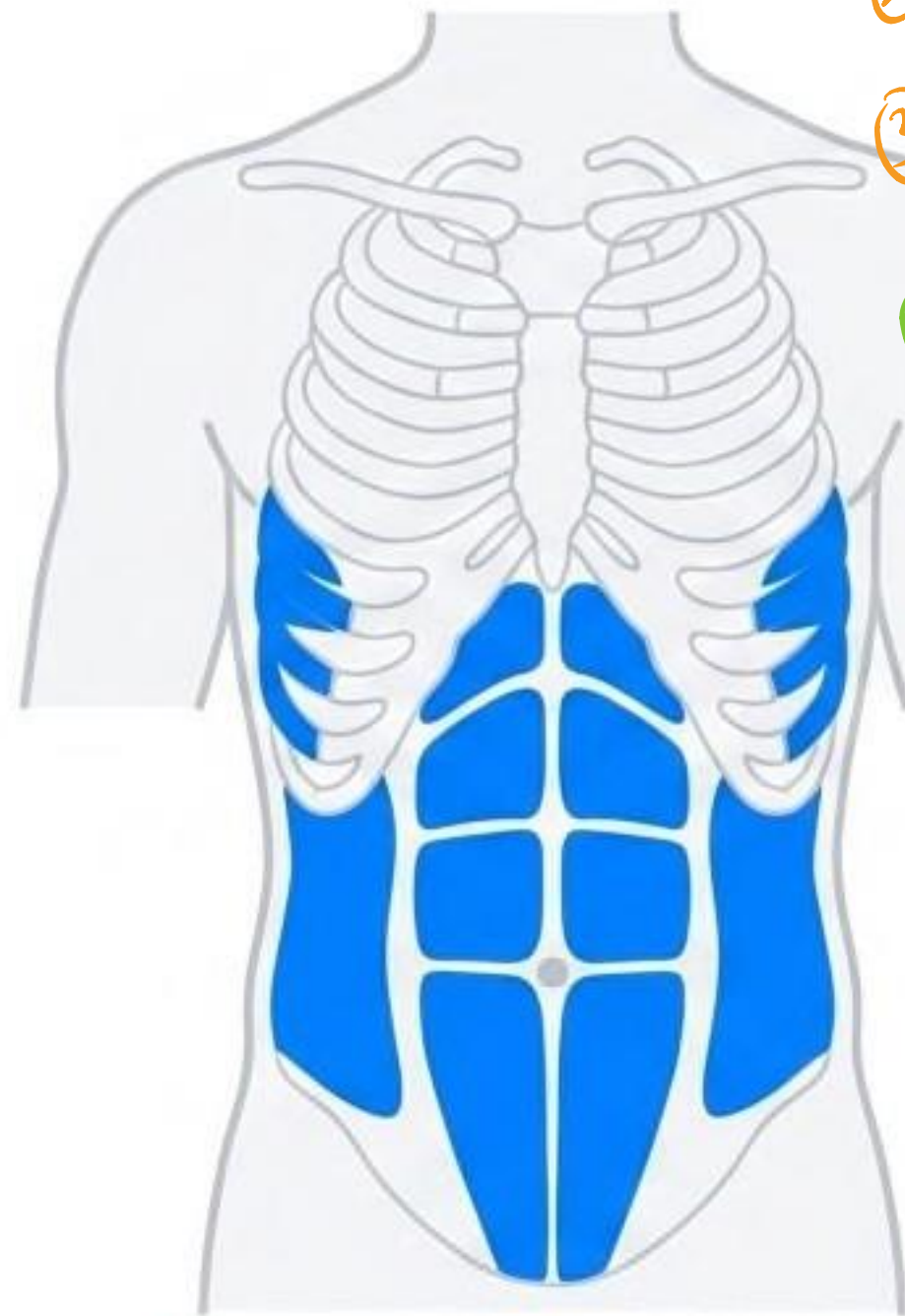
- Purely **Passive** elastic recoil of the lungs and chest wall.
 - Alveolar pressure rises to **+1 cmH₂O**, driving air out until pressure equilibrates at 0.

Onset of Exhalation

Forced Exhalation

- Active muscle recruitment actively compresses the thorax.
- Required for maximum expiration below end-tidal resting level.

✳ Muscles of Expiration: Active Thoracic Compression



① 腹直肌

② 腹内/外斜肌

③ 腹横肌

④ 前锯肌

⑤ 後锯肌

⑥ 闊背肌

Active Exhalation Group

- Rectus abdominis, External/Internal oblique, Transverse abdominal.
- Serratus (anterior, posterior) and Latissimus dorsi.

Mechanics:

Actively compress the thoracic cavity to forcefully drive alveolar pressure above atmospheric levels.

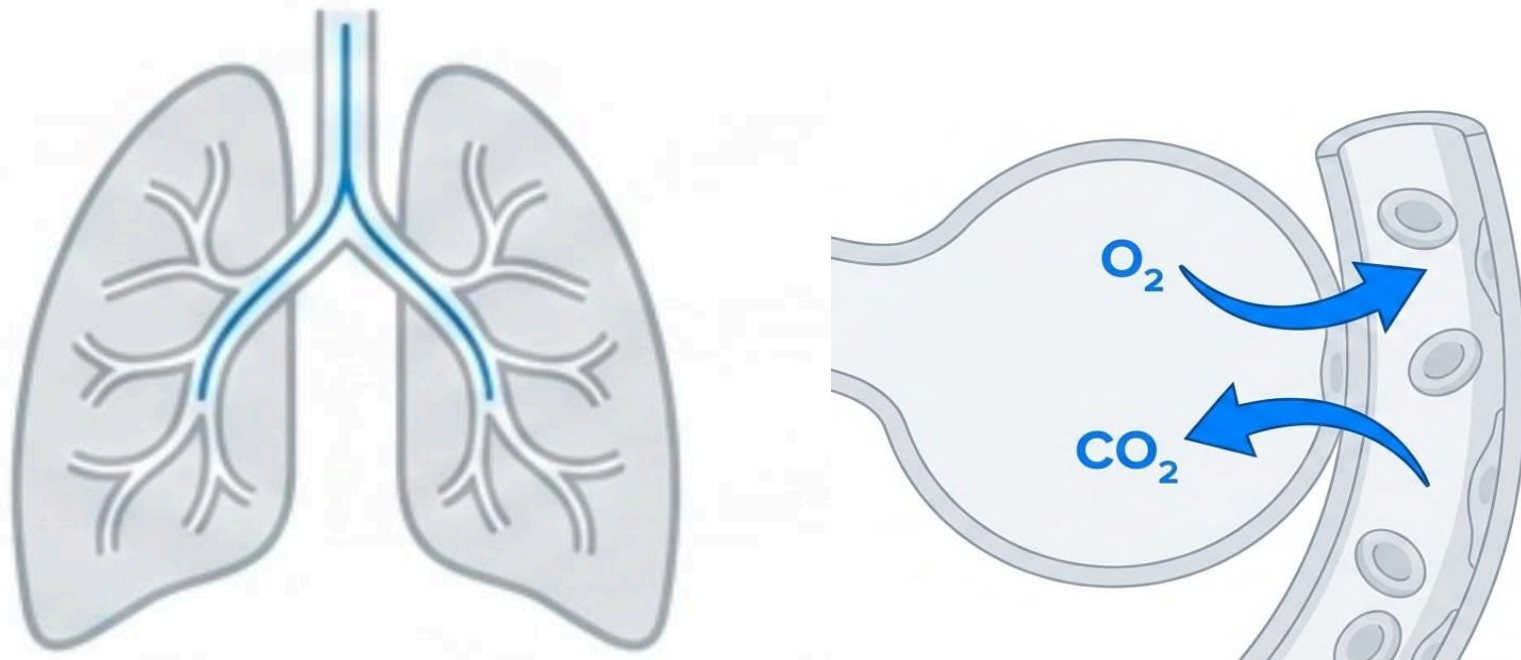
The Dual Components of Respiration

1 ATM = 760 mmHg

External Respiration

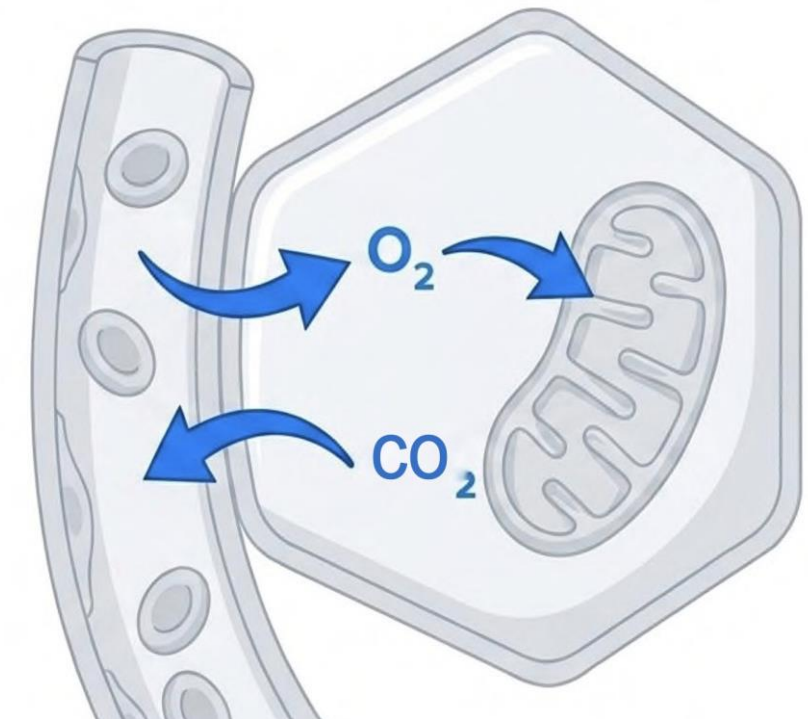
Occurs at the pulmonary [☆]A-C membrane. The interface between the environment and the bloodstream.

O₂ 治療 → 增加氧分壓



Internal Respiration

Occurs at the systemic capillaries and tissue cells. The interface between the bloodstream and mitochondria.



Ventilation vs. Respiration

	VENTILATION	RESPIRATION
Core Mechanism	Bulk gas flow	Molecular diffusion
Location	Conductive Airways	A-C Membrane & Cellular Tissues
Clinical Units	L/min or mL	mmHg or Torr
Clinical Support	Mechanical Ventilator/Ambu	Oxygen Therapy/ECMO

Standardized Measurement Metrics

Ventilator Pressures

Measured primarily in centimeters of water (cmH₂O) or millibars (mbar).

Input gas source pressure is displayed in psig.

Blood & Gas Tensions

Measured in millimeters of mercury (mmHg) or Torr.

International System (SI)

The kilopascal (kPa) serves as the universal scientific standard.

Translating Between Clinical Measurements

$$1 \text{ ATM} = 760 \text{ mmHg} = 1034 \text{ cmH}_2\text{O}$$

$$1 \text{ mmHg} = 1.36 \text{ cmH}_2\text{O}$$

$$1 \text{ kPa} = 7.5 \text{ mmHg}$$

$$1 \text{ cmH}_2\text{O} = 0.735 \text{ mmHg} = 0.98 \text{ mbar} = 0.0142 \text{ lb/in}^2 \text{ (psig)}$$

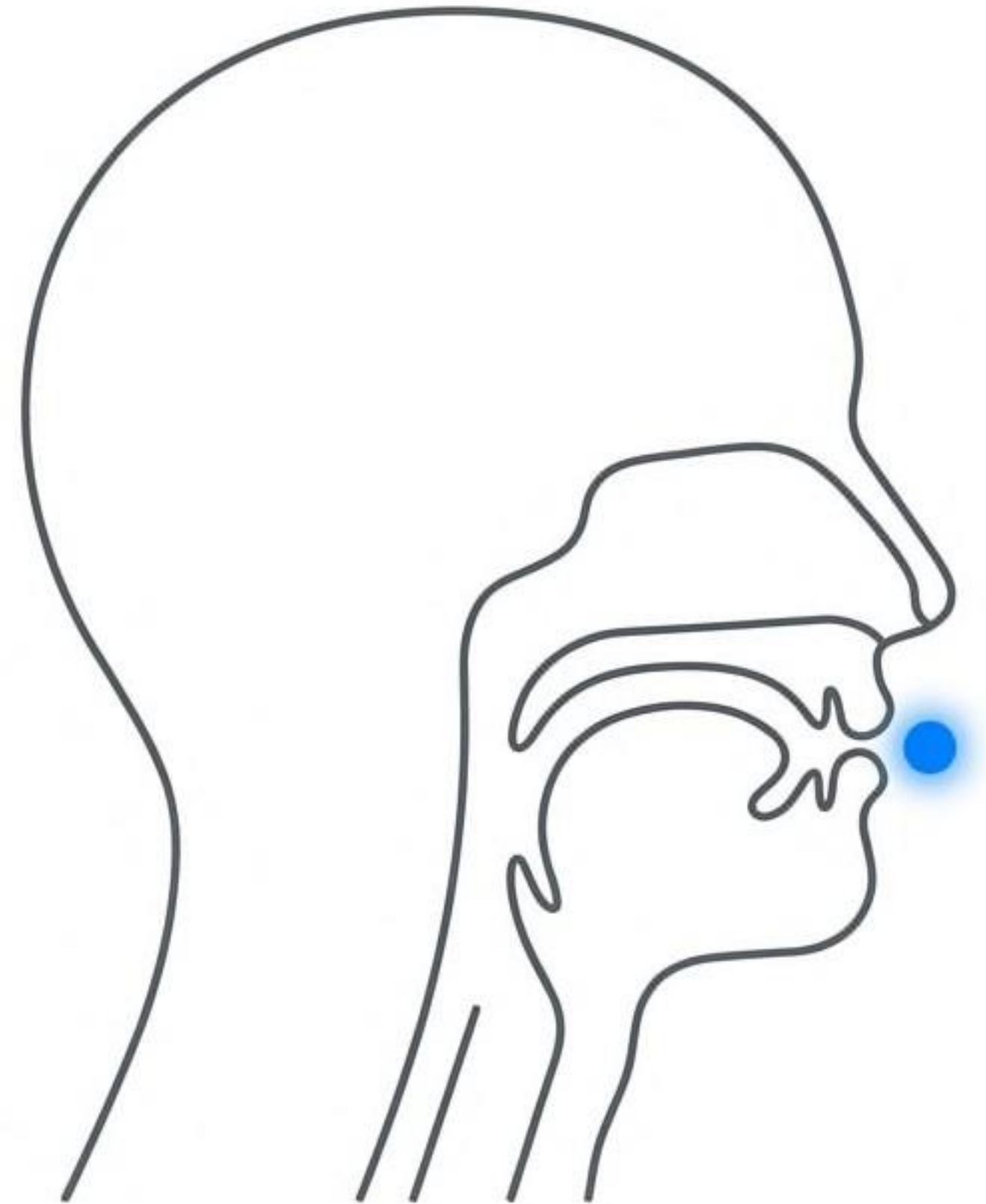
(Relative pressure / Gauge pressure)

Airway Opening Pressure (P_{awo})

Terminology: Also referred to as **mouth pressure (PM)**, airway pressure (P_{aw}), upper airway pressure, mask pressure, or proximal airway pressure.

Baseline Dynamics: P_{awo} is inherently equal to **0 cmH₂O** during unassisted, **spontaneous breathing**.

Clinical Shift: This value strictly changes only if external pressure is mechanically applied to the upper airway.



Body Surface Pressure (**Pbs**)

- Defines the pressure exerted on the outer surface of the entire chest wall.
- Normally registers statically at 0 cmH₂O (equal to atmospheric pressure).

Clinical Exceptions

- ^{高壓氧}Hyperbaric Chamber: Forces Pbs into positive values.
- ^{鐵肺 / 負壓通氣}Iron Lung / Negative Pressure Ventilator: Forces Pbs into subatmospheric (negative) values.

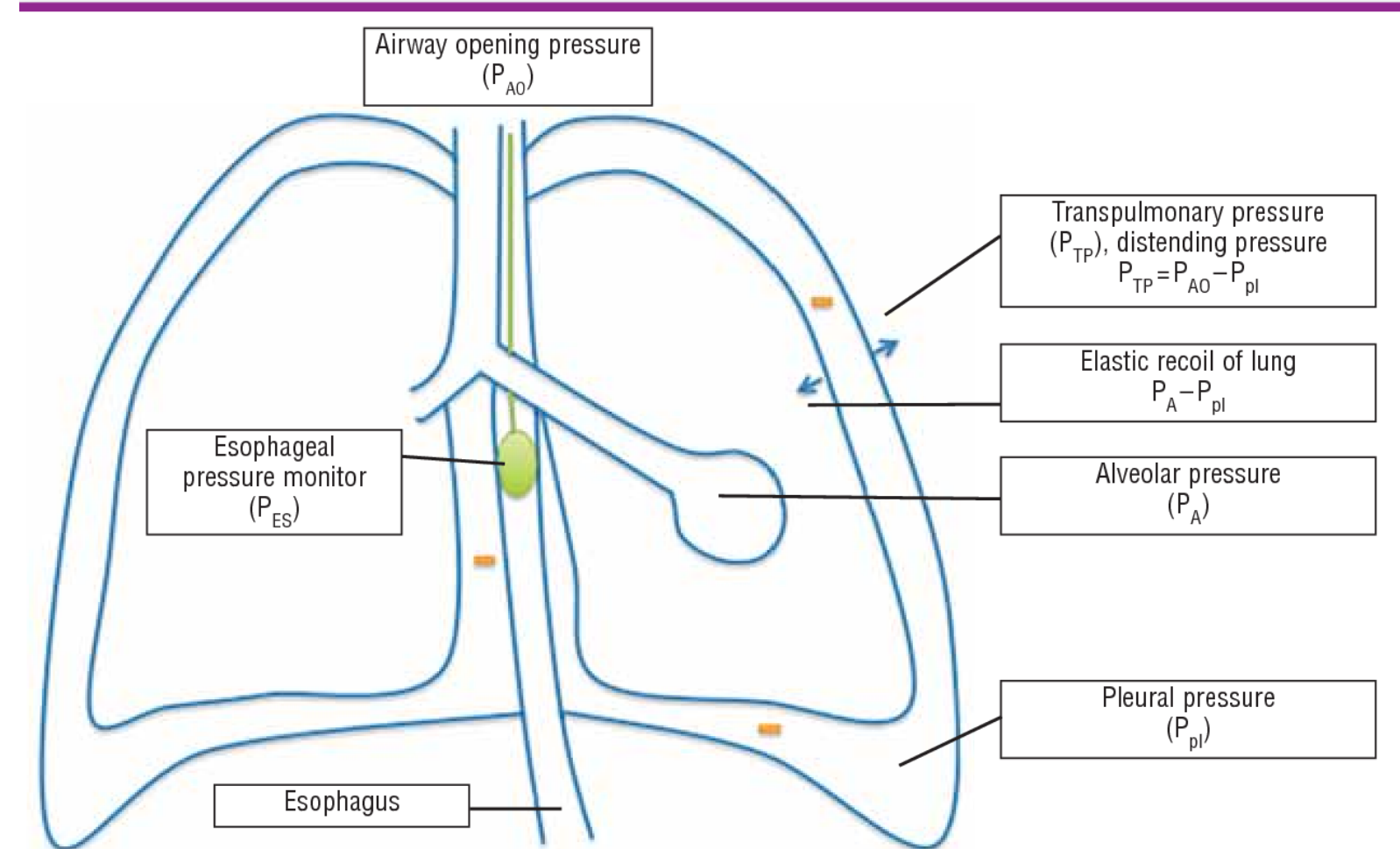
Intrapleural Pressure (P_{pl})

The measurement of pressure within the potential space between the parietal and visceral pleurae.

Dynamic Changes:

- 21 - End of Expiration: Normally ~ -5 cmH₂O.
- 0.2 - End of Inspiration: Drops to ~ -10 cmH₂O as thoracic cavity expands.

Clinical Surrogate: Direct P_{pl} measurement is highly invasive; esophageal pressure (P_{es}) is utilized as its standard clinical surrogate.

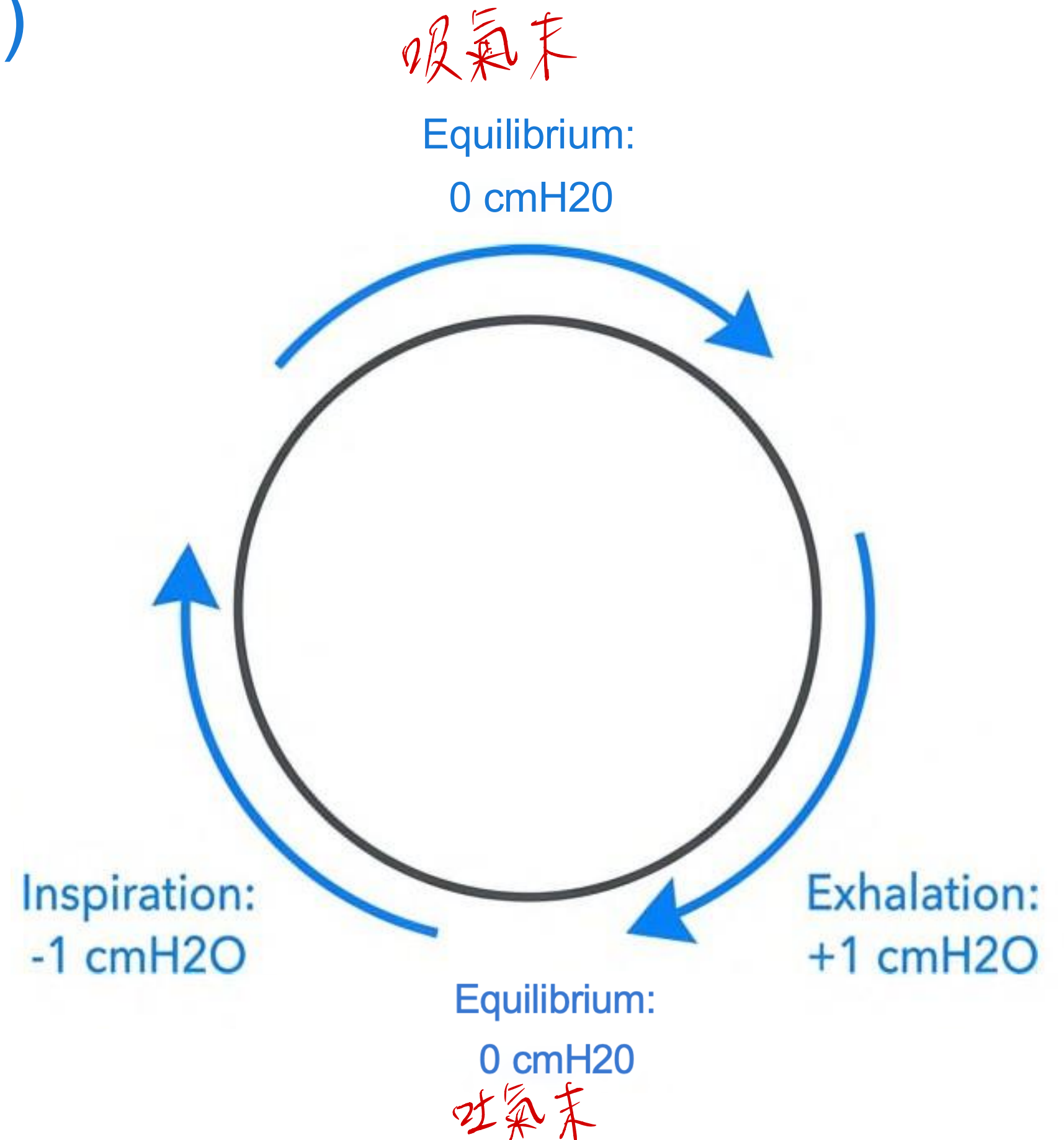


Alveolar Pressure (PA / Palv)

Alternatively known as intrapulmonary pressure or lung pressure.

Spontaneous Breathing Cycle:

- **Inspiration:** Drops to -1 cmH₂O (can range safely from -3 to -5 cmH₂O).
- **Exhalation:** Rises to +1 cmH₂O.
- **Equilibrium:** Reaches exactly 0 cmH₂O at both end-inspiration and end-expiration.



Mapping the Four Basic Pressure Gradients

Pressure gradients describe the vital mathematical differences between key anatomical points, directly driving physical ventilation.

Transrespiratory Pressure

PTR

Transairway Pressure

PTA

肺泡-肋膜

Transpulmonary Pressure

PL / PTP

Transthoracic Pressure

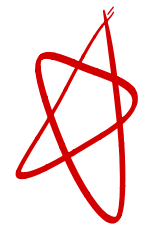
PW / PTT

Transairway Pressure (PTA): Overcoming Resistance

$$0.5 = R_{aw} \times Flow (L/s)$$

$$PTA = P_{awo} - P_{alv}$$

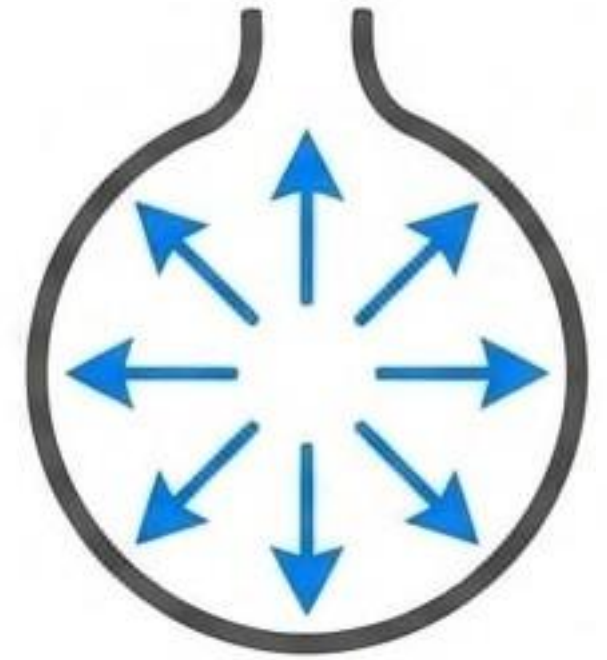
- Represents the precise pressure gradient required to produce airflow in the conductive airways and overcome mechanical airway resistance.



- Clinical Correlation: Equal to Peak Inspiratory Pressure (PIP) minus Plateau Pressure (P_{plateau}).
- Baseline: Normal PTA at a flow rate of 0.5 L/s is approximately 0.6 cmH₂O.

Transpulmonary Pressure (PL): Alveolar Distention

$$PL = P_{alv} - P_{pl}$$



- Responsible for maintaining internal alveolar inflation, frequently referred to as the **alveolar distending pressure**.
- Critical Rule: **PL must remain a positive** value at all times to keep the alveoli open and definitively prevent alveolar collapse (atelectasis).

Traditional Transpulmonary Pressure (PTP)

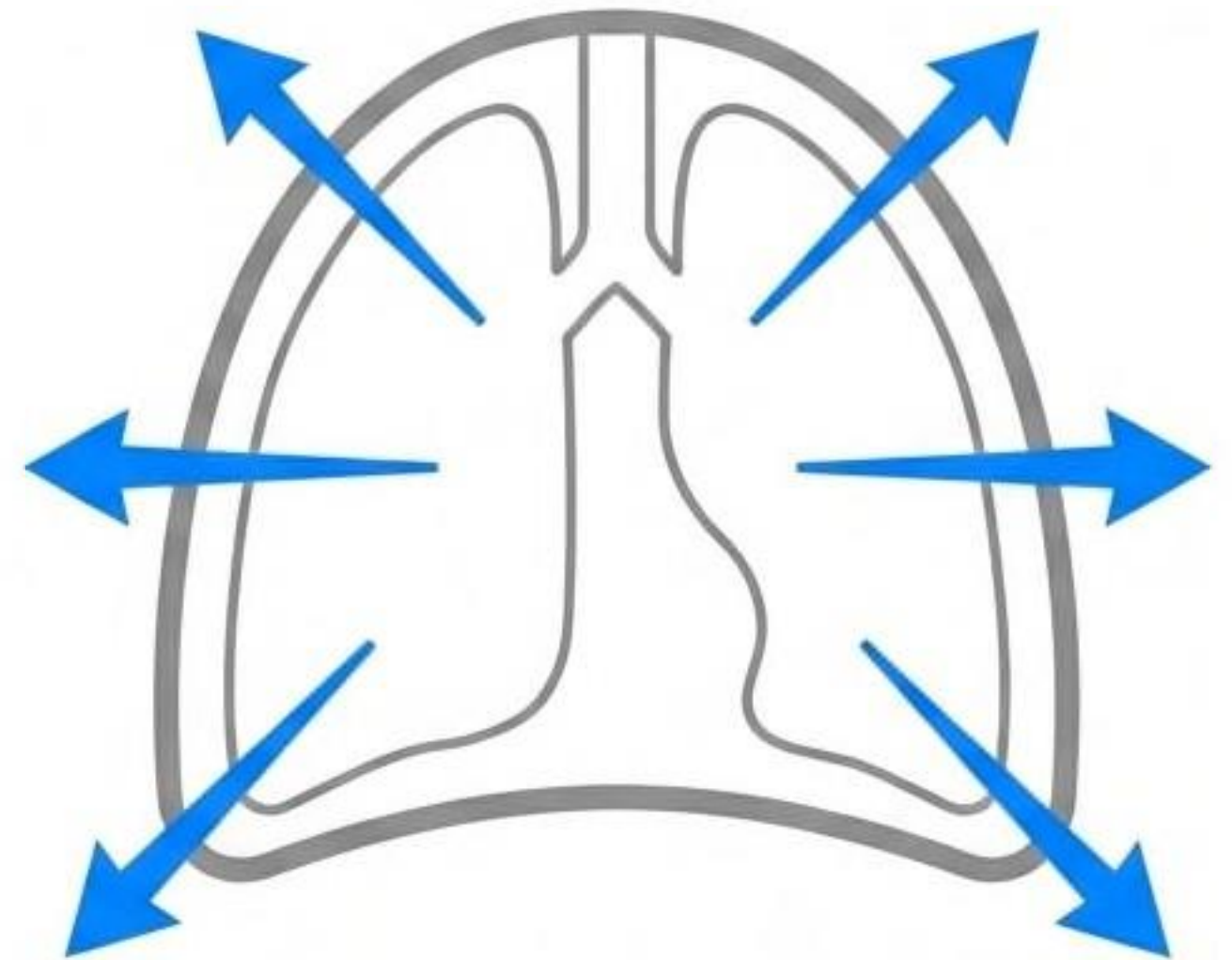
$$PTP = \overset{\text{Air way opening pressure}}{\underline{P_{awo}}} - P_{pl}$$

- Defines the pressure across the entire lung system required to distend it (including airways).
- Static Reality: Under zero-flow static conditions, PTP is mathematically identical to PL.
- Clinical Application:
 - Positive Pressure Ventilation (PPV) artificially increases PTP by driving up P_{awo} .
 - Negative Pressure Ventilation (NPV) artificially increases PTP by driving down P_{pl} .

Transthoracic Pressure (PW / PTT): Expanding the Chest Wall

$$PW = P_{alv} - P_{bs}$$

- Defines the exact pressure gradient between the internal alveolar space and the external body surface.
- Represents the total pressure required to physically expand or contract both the elastic lungs and the heavy chest wall simultaneously.



Transrespiratory Pressure (PTR): Total Inflation Pressure

正壓通氣用

$$\text{PTR} = \text{Pawo} - \text{Pbs}$$

Represents the absolute total inflation pressure read on the mechanical ventilator gauge during PPV (assuming Pbs remains at 0).

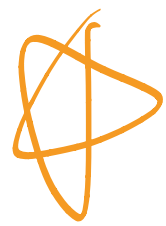
$$\text{PTR} = \text{PTT} + \text{PTA}$$



Transthoracic
elastic load

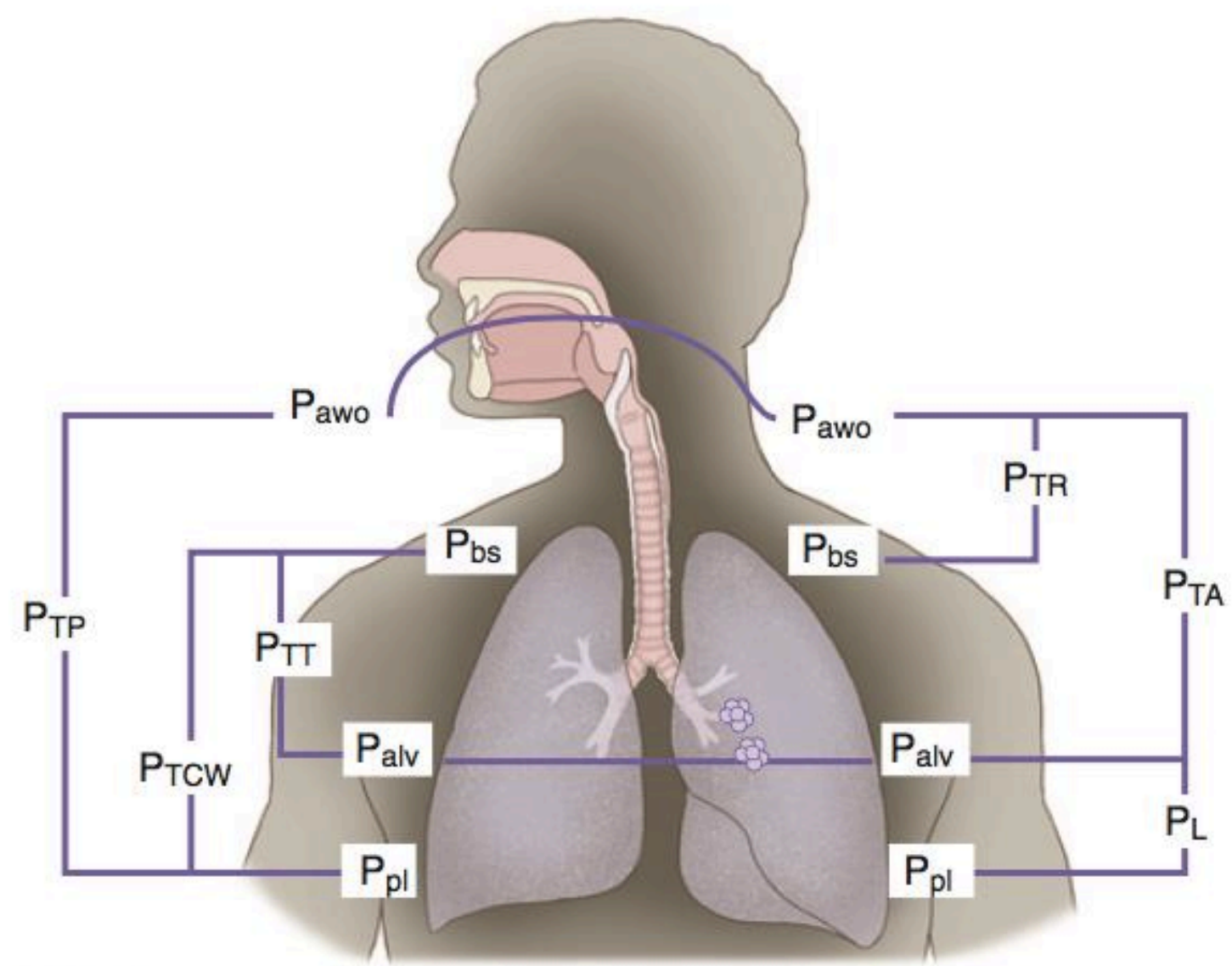


Transairway
resistive load



Pressure and Pressure Gradients

$P_{AO_2} > P_{aO_2} > P_{vO_2}$



P_{awo} - Mouth or airway opening pressure	
P_{bs} - Body surface pressure	
P_{alv} - Alveolar pressure	
P_{pl} - Pleural pressure	
Transrespiratory pressure - P_{TR}	$= P_{awo} - P_{bs}$
Transairway pressure - P_{TA}	$= P_{awo} - P_{alv}$
Transpulmonary pressure - P_{TP}	$= P_{awo} - P_{pl}$
Transalveolar pressure - P_L	$= P_{alv} - P_{pl}$
Trans-chest wall pressure - P_{TCW}	$= P_{pl} - P_{bs}$
Transthoracic pressure - P_{TT}	$= P_{alv} - P_{bs}$

很重要

FIGURE 46-1 Pressures and pressure gradients in the lung. Airflow is a function of the transairway pressure (P_{ta}), which is the pressure gradient between the airway (P_{awo}) and the alveoli (P_{alv}). Transalveolar pressure (P_L) maintains alveolar inflation, and transpulmonary pressure (P_{tp}) is the pressure needed to expand the lungs and chest wall.

Transdiaphragmatic & Transmural Boundaries

Block 1

Transdiaphragmatic Pressure (Pdi)

$$P_{di} = P_{pl} - P_{ab}$$

The exact gradient between pleural and abdominal pressures; vital for the assessment of active diaphragmatic movement and muscle function.

Block 2

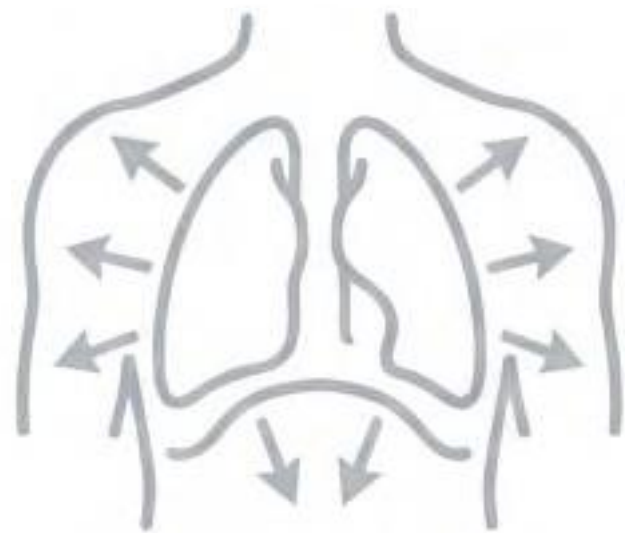
Transmural Pressure Principle

A universal physical principle defining the pressure difference occurring strictly across any anatomical wall or separator.

- Examples in System: $P_{pl} - P_{bs}$ acts across the heavy chest wall; PL acts directly as the transalveolar separator.

Types of Mechanical Ventilation

The mechanical ventilator provides the active physical energy required to alter pressure gradients, overcome elastance and resistance, and expand the thorax [Egan's Chap, 47, p. 1018].



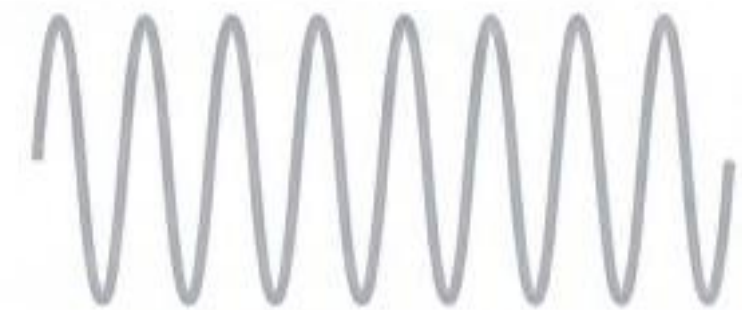
Negative Pressure Ventilation
(NPV)

Decreases pleural pressure by exposing the chest to subatmospheric pressure.



Positive Pressure Ventilation
(PPV)

Increases airway opening pressure to actively blow gas into the lungs.



High-Frequency Ventilation
(HFV)

Utilizes above-normal ventilating rates with strictly below-normal tidal volumes.

Negative Pressure Ventilation (NPV)

Mimicking normal physiological breathing mechanics to preserve natural intrathoracic pressure changes [Pilbeam Chap. 1, p. 10].

Physiological Mimicry:

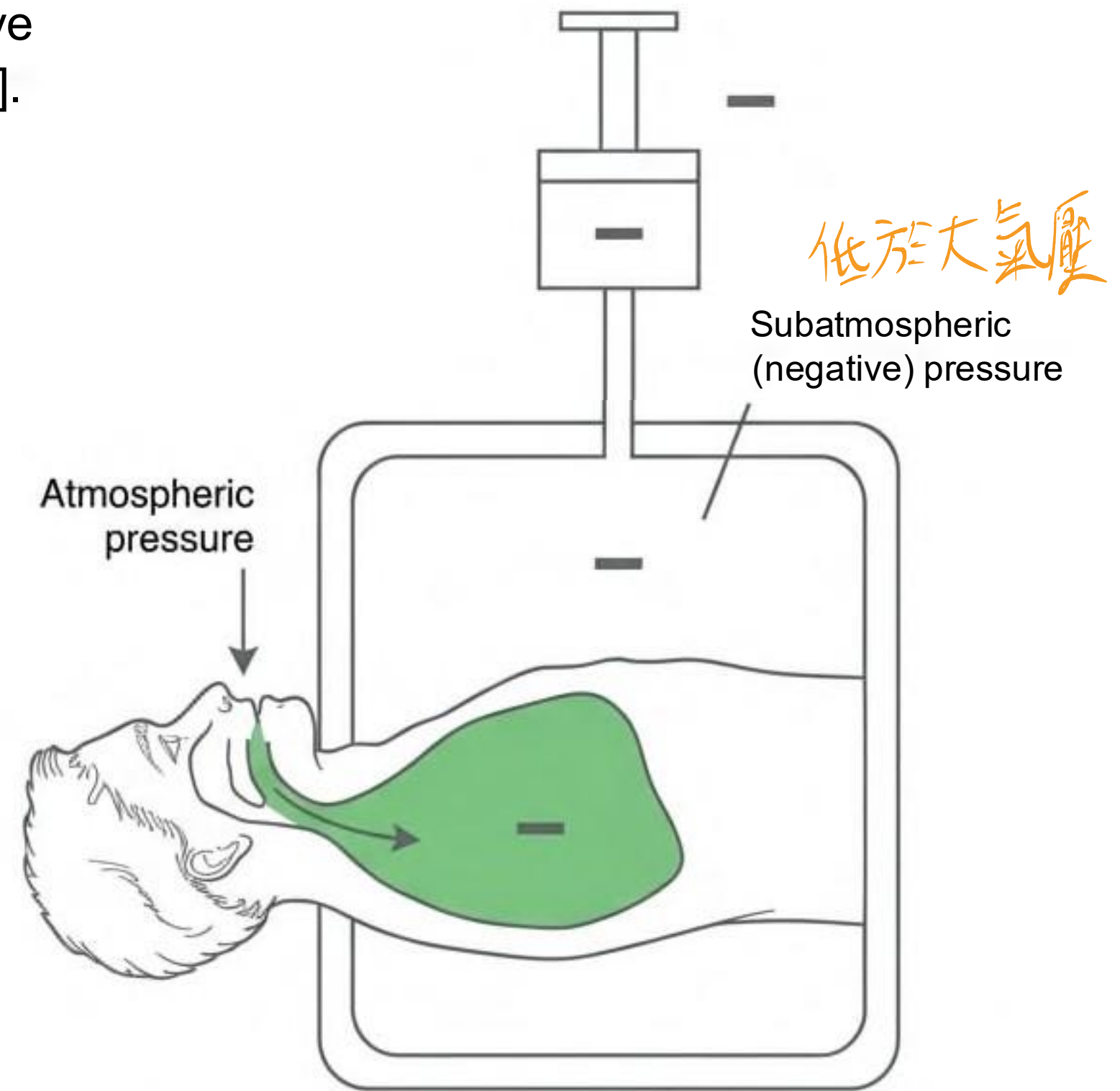
NPV attempts to replicate the function of the respiratory muscles, preserving normal physiological intrathoracic pressure changes.

Chamber Enclosure:

The patient's thorax and body are enclosed in an “**airtight chamber**” or cuirass, while the head and neck remain completely exposed to ambient atmospheric pressure.

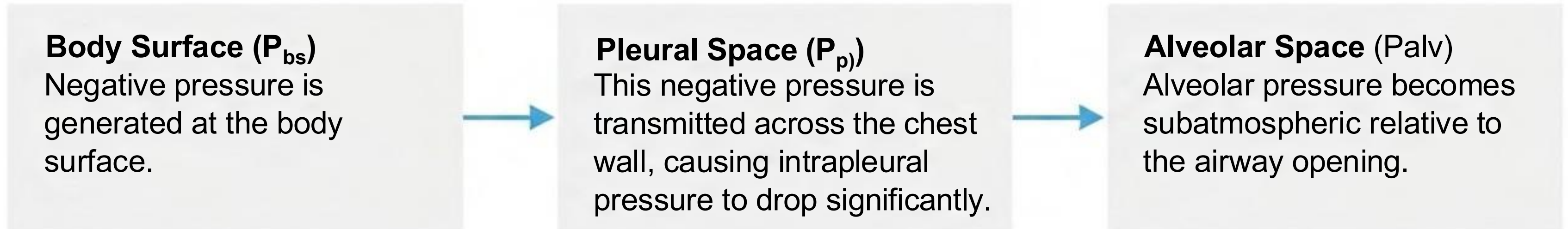
Pressure Generation:

A “**subatmospheric pressure**” is actively generated around the thoracic area to initiate the inspiratory phase.



NPV Mechanics and Pressure Gradients

How subatmospheric pressure translates into mechanical lung expansion.



The Resulting Gradients:

- **Transrespiratory Pressure (P_{TR}):** Increases as the gradient between the airway opening and the negative body surface widens.
- **Transpulmonary Pressure (P_L):** Increases as P_p becomes increasingly negative, expanding the alveoli to draw ambient air into the lungs.

$$P_L = P_{alv} - P_{pl}$$

Advantages of NPV

Clinical benefits of preserving natural pressure mechanics.



Airway Preservation

Maintains upper airway function without the requirement for an artificial airway, such as an endotracheal or tracheostomy tube.



Patient Autonomy

Because the head is exposed to ambient air, conscious patients can easily eat, drink, and talk while being continuously ventilated.



Cardiovascular Stability

Produces minimal physiological cardiovascular side effects in normovolemic patients compared to positive pressure modes.

Disadvantages and Complications of NPV

Anatomical and physiological limitations of subatmospheric chamber ventilation.

Cardiovascular Pooling (“Tank Shock”) 血液留在腹部

In ^{低血容積} hypovolemic patients, subatmospheric pressure on the lower body causes significant blood pooling in the abdomen, severely reducing venous return to the heart.

Airway Obstruction Risks

OSA. 聲門關閉

High risk of glottis closure and the development of Obstructive Sleep Apnea (OSA), particularly in patients with neuromuscular dysfunction.

Patient Accessibility

The physical chamber severely limits medical access to the patient's body for routine nursing care or emergent interventions.

Positive Pressure Ventilation (PPV)

endo

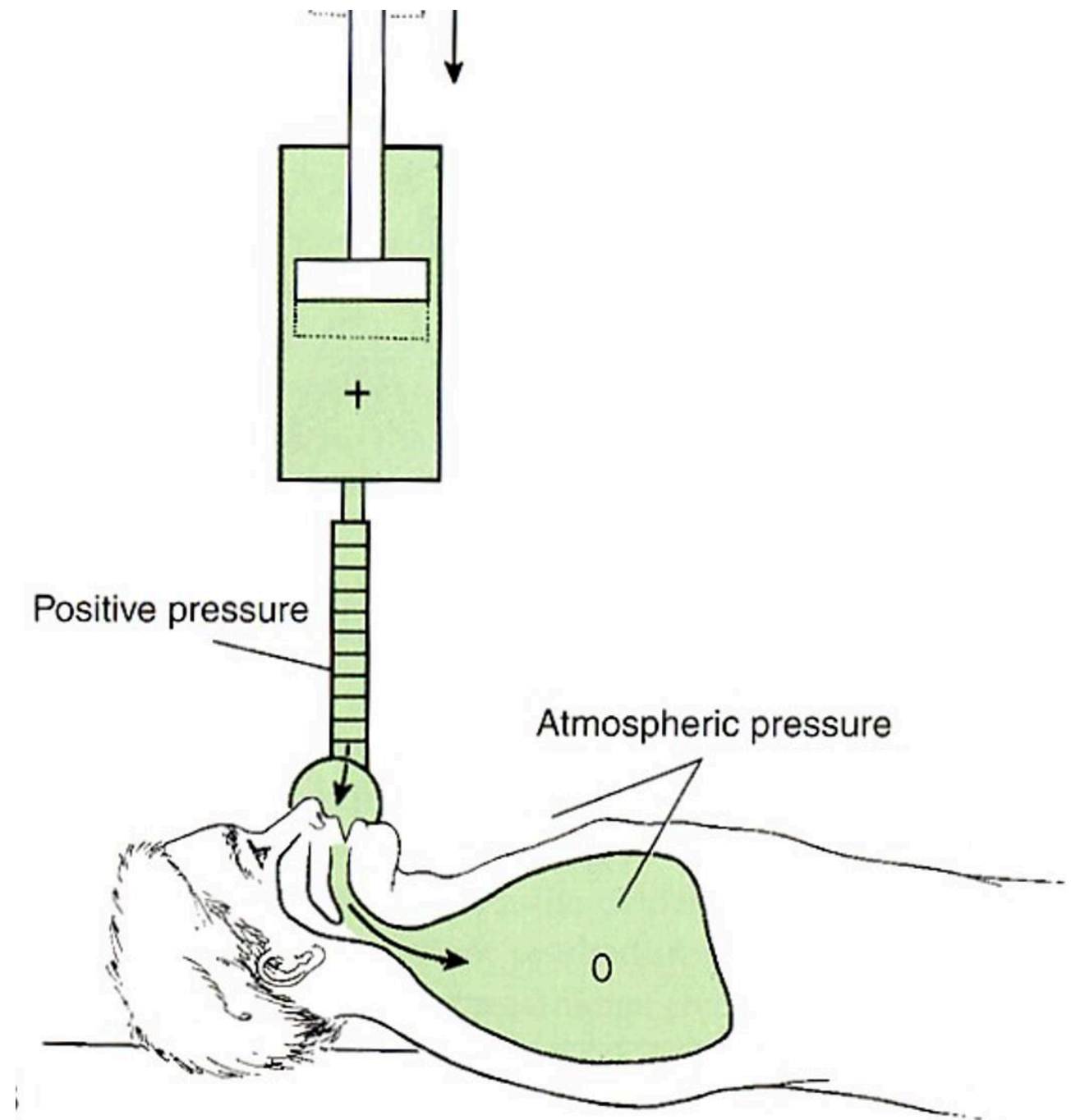
The current clinical standard for active mechanical inflation.

Active Inflation: The mechanical ventilator actively blows air into the patient's lungs, overriding or replacing the natural muscular effort of the diaphragm [P.

Patient Interface: Delivered directly via an endotracheal tube (invasive) or a positive pressure mask (noninvasive).

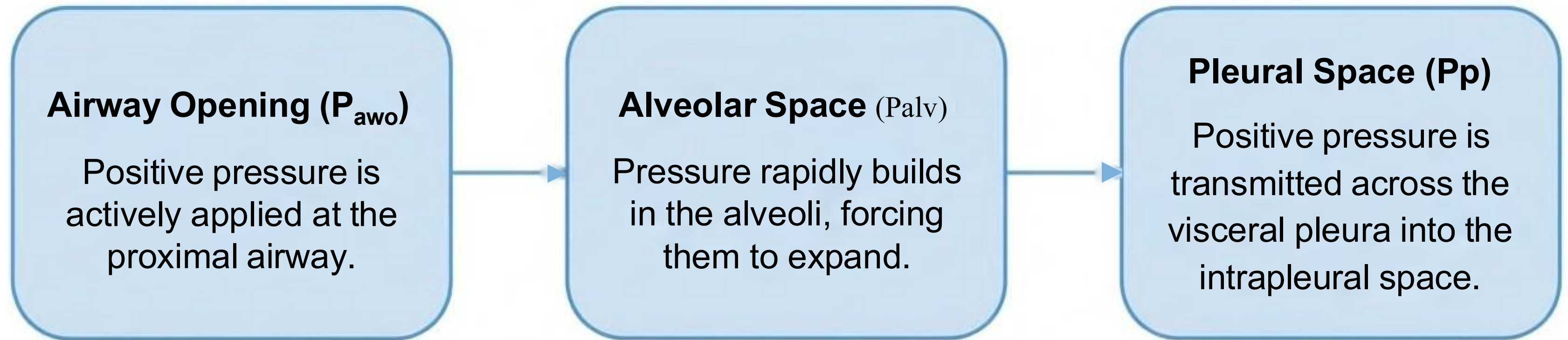
The Inspiratory Gradient: Establishes gas flow by raising the proximal airway opening pressure (P_{awo}) significantly above the alveolar pressure (P_a).

$P_{awo} \uparrow$



PPV Mechanics and Pressure Transmission

Reversing natural spontaneous mechanics to achieve lung inflation



The Mechanical Reversal:

Unlike normal physiology, intrapleural pressure (P_p) actually becomes "**positive at the end of inspiration**". The inflation pressure must overcome both the elastance of the lung/chest wall and the resistance of the airways.

TABLE 1-2

Changes in Transpulmonary Pressure* Under Varying Conditions

Passive Spontaneous Ventilation Pressure	End Expiration	End Inspiration
Intraalveolar (intrapulmonary)	0 cm H ₂ O	0 cm H ₂ O
Intrapleural	−5 cm H ₂ O	−10 cm H ₂ O
Transpulmonary	$P_L = 0 - (-5) = +5$ cm H ₂ O	$P_L = 0 - (-10) = 10$ cm H ₂ O
Negative Pressure Ventilation		
Intraalveolar (intrapulmonary)	0 cm H ₂ O	0 cm H ₂ O
Intrapleural	−5 cm H ₂ O	−10 cm H ₂ O
Transpulmonary	$P_L = 0 - (-5) = +5$ cm H ₂ O	$P_L = 0 - (-10) = 10$ cm H ₂ O
Positive Pressure Ventilation		
Intraalveolar (intrapulmonary)	0 cm H ₂ O	9-12 cm H ₂ O [†]
Intrapleural	−5 cm H ₂ O	2-5 cm H ₂ O [†]
Transpulmonary	$P_L = 0 - (-5) = +5$ cm H ₂ O	$P_L = 10 - (2) = +8$ cm H ₂ O [†]

* $P_L = P_{alv} - P_{pl}$.

†Applied pressure is +15 cm H₂O.

The Unifying Principle: Regardless of whether gas is pulled in (Spontaneous/NPV) or pushed in (PPV), all three modes successfully expand **transpulmonary pressure (P_L)** to inflate the alveoli

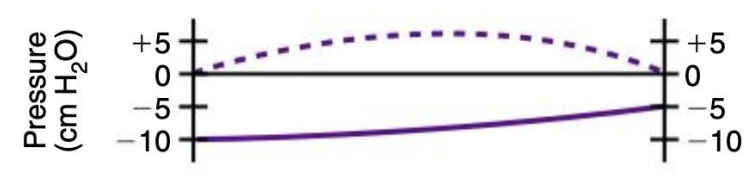
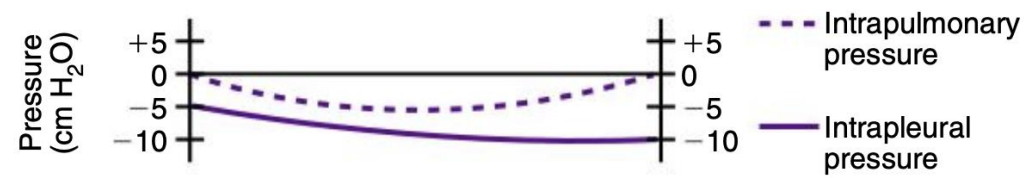
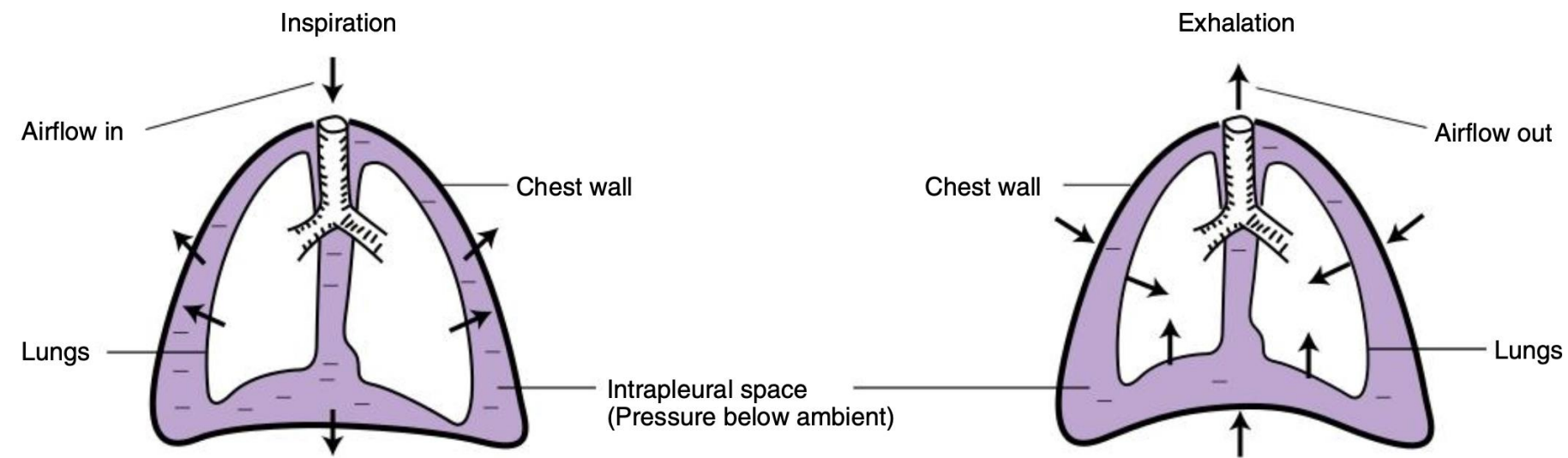
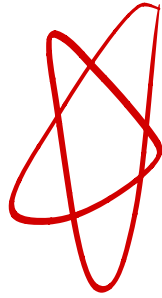
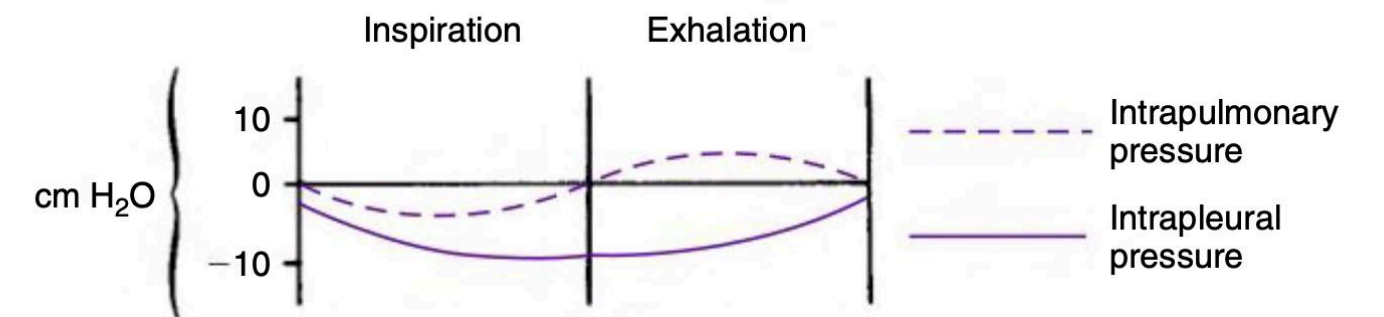
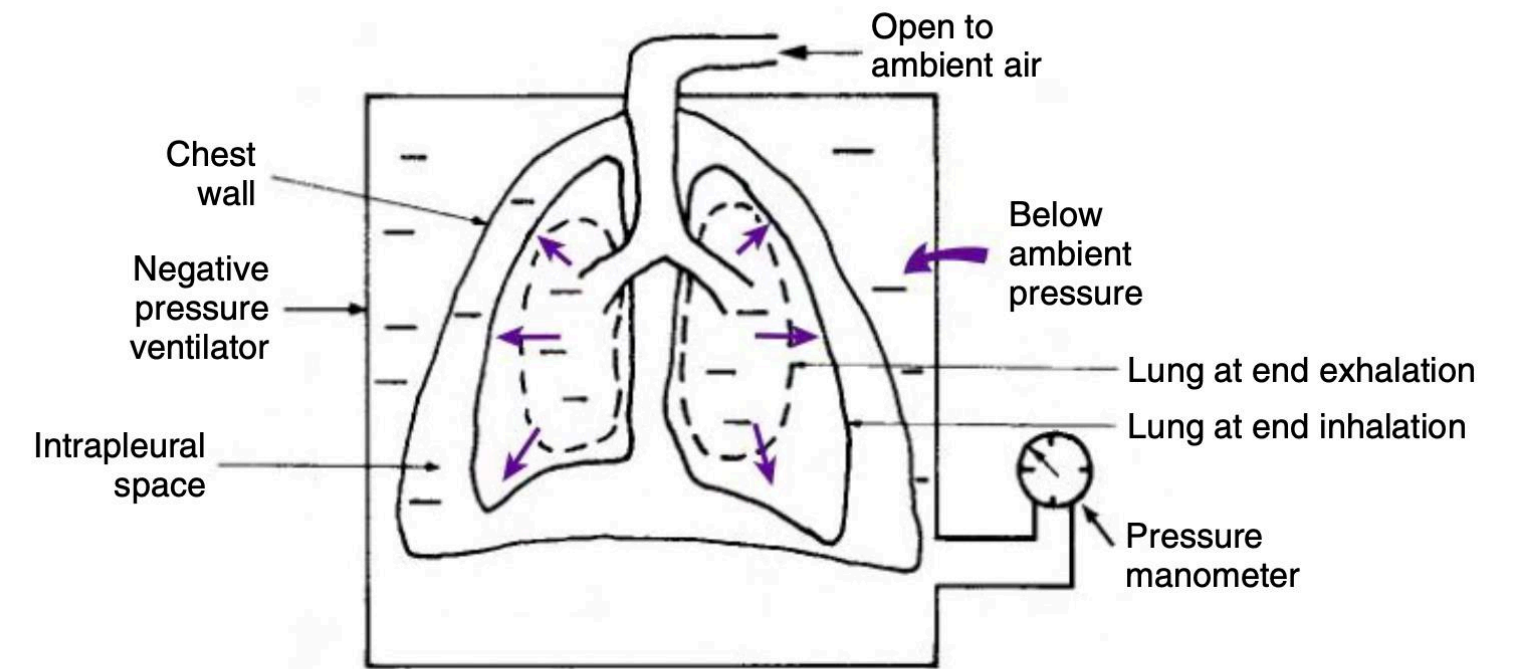


Fig. 1.2 The mechanics of spontaneous ventilation and the resulting pressure waves (approximately normal values). During inspiration, intrapleural pressure (P_{pl}) decreases to -10 cm H₂O. During exhalation, P_{pl} increases from -10 to -5 cm H₂O. (See the text for further description.)

Spontaneous Breathing



NPV

想像在一個瓶子，
上面套氣球，再往
裡面通氣，氣球變大，
內部壓力上升，氣球跟
瓶子之間的壓力也上升

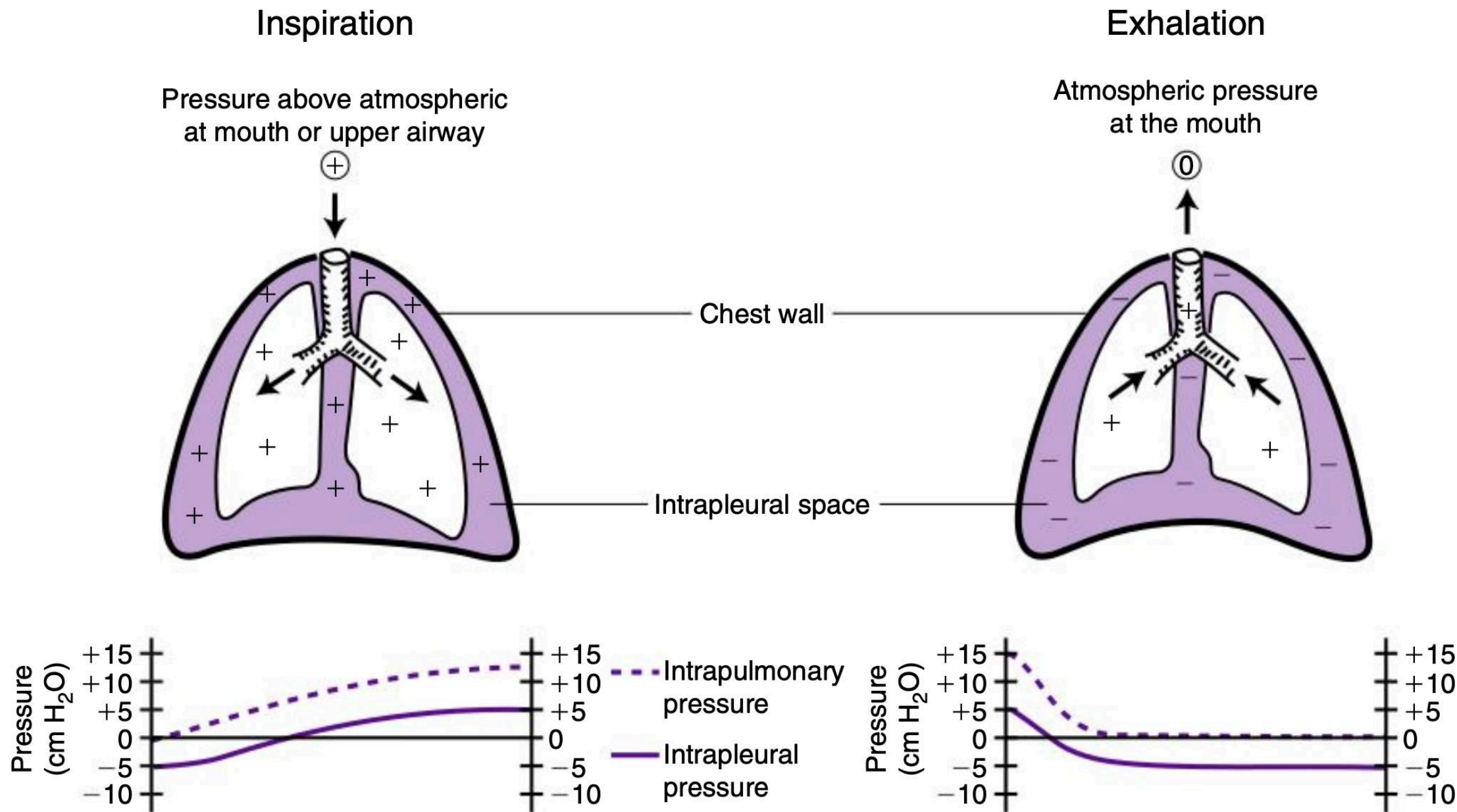
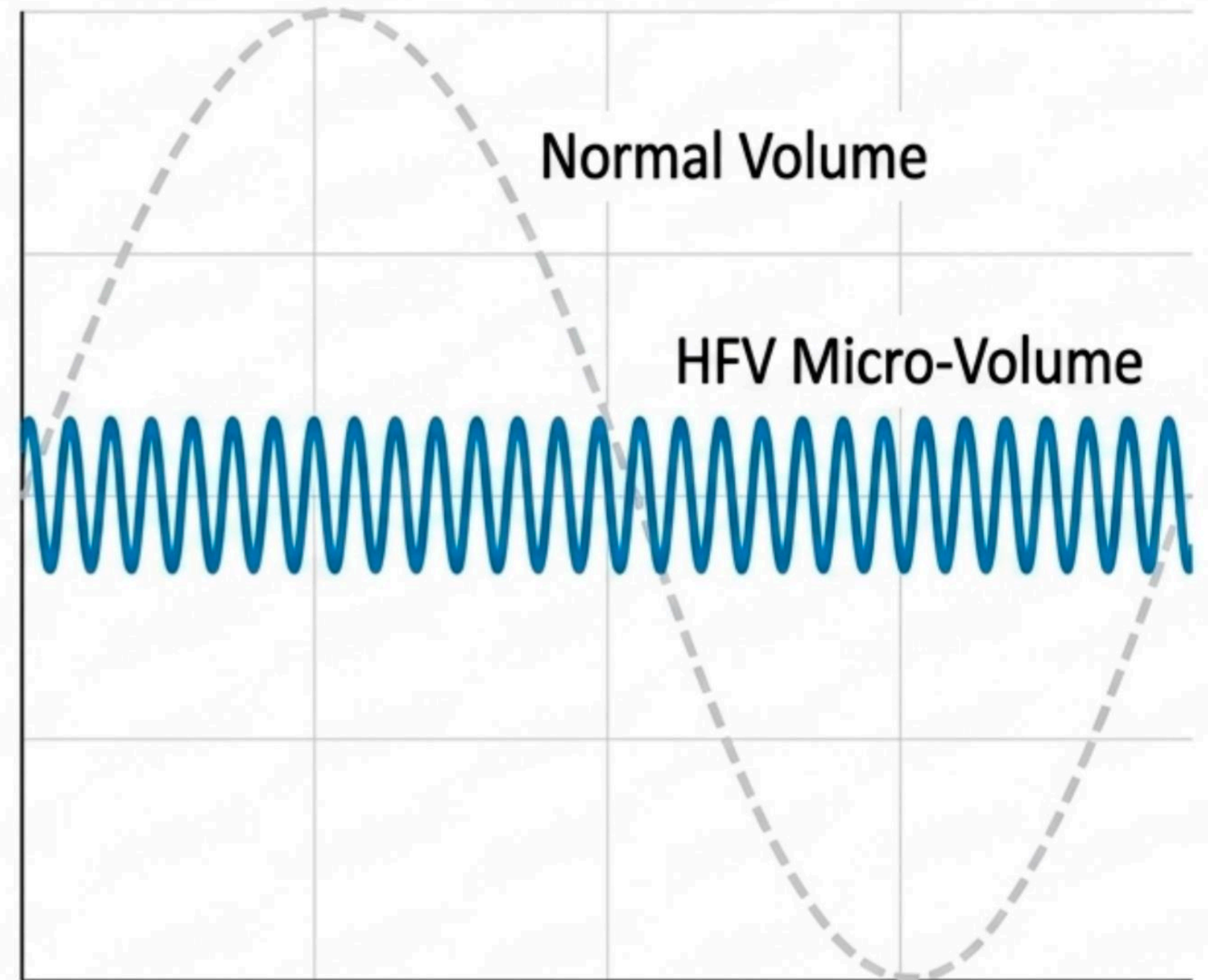


Fig. 1.8 Mechanics and pressure waves associated with positive pressure ventilation. During inspiration, as the upper airway pressure rises to about +15 cm H₂O (not shown), the alveolar (intrapulmonary) pressure is zero; as a result, air flows into the lungs until the alveolar pressure rises to about +9 to +12 cm H₂O. The intrapleural pressure rises from about 5 cm H₂O before inspiration to about +5 cm H₂O at the end of inspiration. Flow stops when the ventilator cycles into exhalation. During exhalation, the upper airway pressure drops to zero as the ventilator stops delivering flow. The alveolar (intrapulmonary) pressure drops from about +9 to +12 cm H₂O to 0 as the chest wall and lung tissue recoil to their normal resting position; as a result, air flows out of the lungs. The intrapleural pressure returns to -5 cm H₂O during exhalation.

High-Frequency Ventilation (HFV)

Utilizing unconventional rates and micro-volumes to preserve gas exchange.

- **The HFV Paradigm:** Utilizes significantly above-normal ventilating rates combined with strictly below-normal tidal volumes.
- **Volume vs. Dead Space:** Tidal volumes delivered are often less than the patients anatomical dead space, relying on alternative gas exchange mechanisms.
- **Clinical Objectives:** Minimize peak airway pressures to prevent barotrauma, preserve efficient gas exchange, and precisely manipulate mean airway pressure (MAP) to optimize oxygenation.



Technical Comparison of HFJV and HFOV

High-Frequency Jet Ventilation (HFJV)

跟 AEM 的
jet 很像

- **Operating Frequency:** 4 to 11 Hz (approx. 240-660 breaths/min).
- **Delivery Mechanism:** Delivers high-velocity gas jets via a specially designed endotracheal tube adaptor or injector.
- **Exhalation Phase:** Completely "passive" exhalation.

as normal spontaneous exhalation
passive exhalation

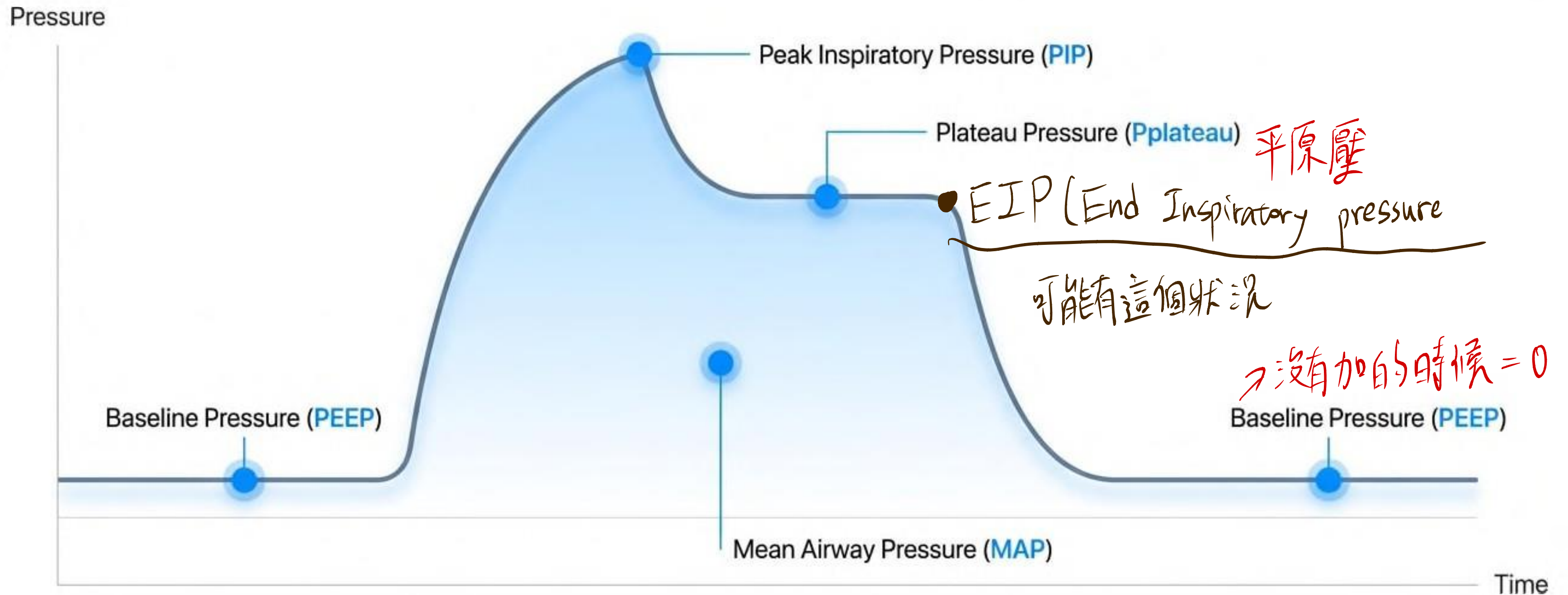
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High-Frequency Oscillatory Ventilation (HFOV)

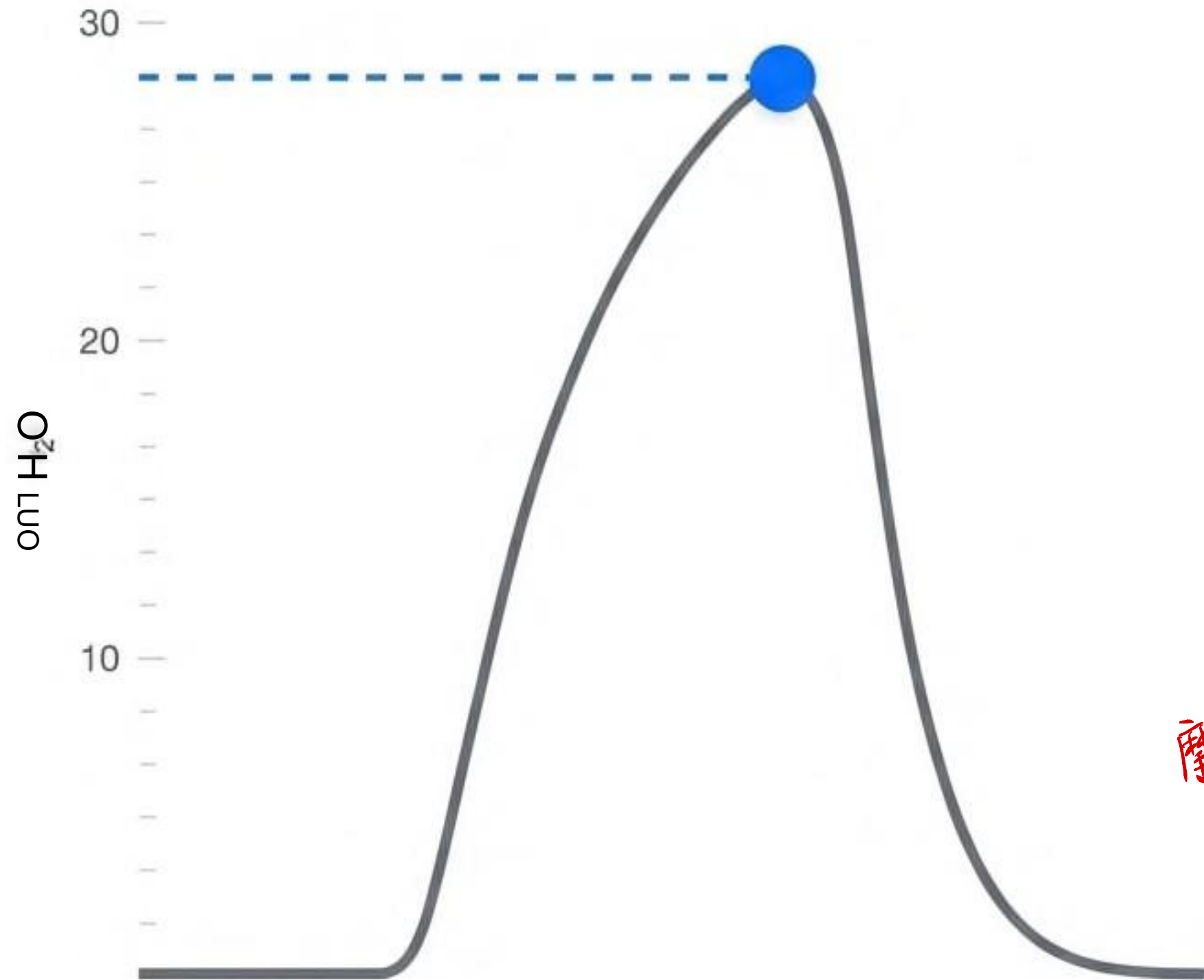
- **Operating Frequency:** 3 to 15 Hz (approx. 180-900 breaths/min).
- **Delivery Mechanism:** Utilizes a piston or stereo speaker device for a rapid to-and-fro gas motion.
- **Exhalation Phase:** Features an "active" exhalation phase, physically pulling gas out of the airway.

Positive Pressure Ventilation (PPV): Core Pressure Parameters

Positive Pressure Ventilation (PPV) artificially blows air into the lungs via a positive pressure gradient, fundamentally altering transrespiratory and transpulmonary pressures.



Peak Inspiratory Pressure (PIP / Ppeak)



Definition

The absolute **highest pressure** recorded at the airway opening at the **end of the inspiratory phase**.

Clinical Function

Represents the total pressure required to force gas through the frictional resistance of the airways **AND** inflate the alveoli with volume.

摩擦阻力

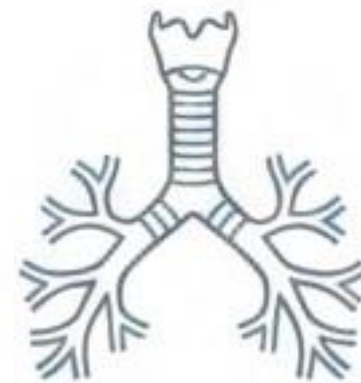
Mathematical Components of PIP

$P_{aw} - P_{alv}$
or
 $R_{aw} \times \text{flow}$

$$PIP = PTA + P_{alv}$$

(at end-inspiration)

Transairway Pressure (PTA)



Overcomes the frictional resistance of the conductive airways (R_{aw}). Represents the pressure lost to airflow dynamics.

Alveolar Pressure (P_{alv})



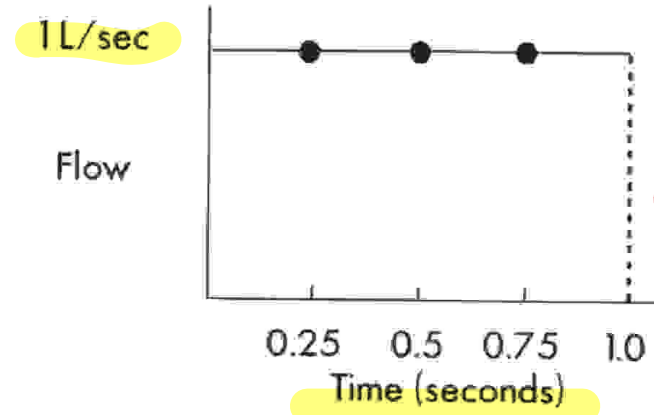
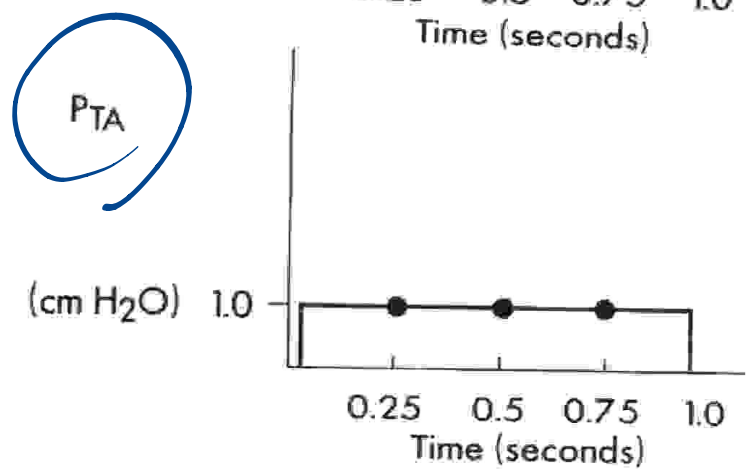
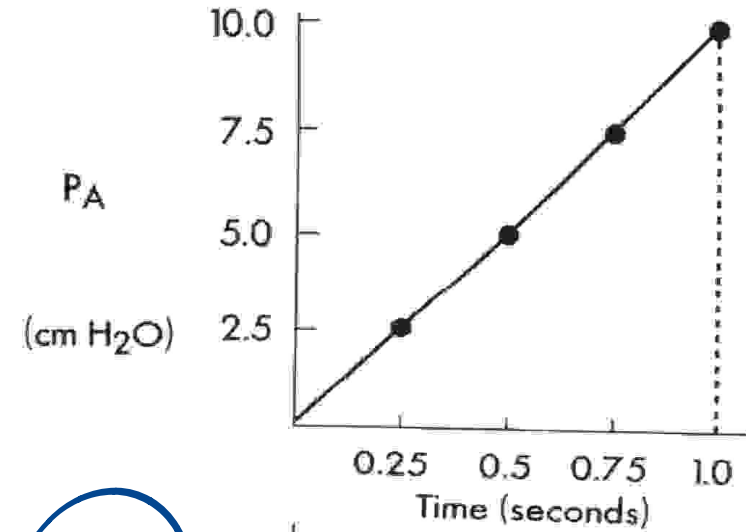
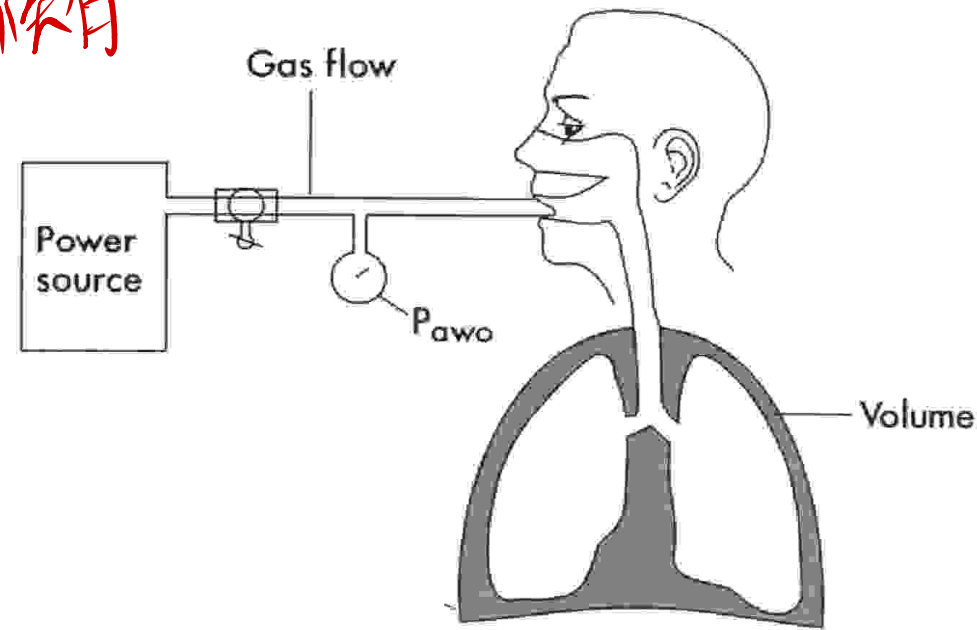
Overcomes the elastic recoil (elastance) of the lungs and chest wall. Represents the pressure required to maintain alveolar inflation.

PTA 只有在有 flow 的時候有

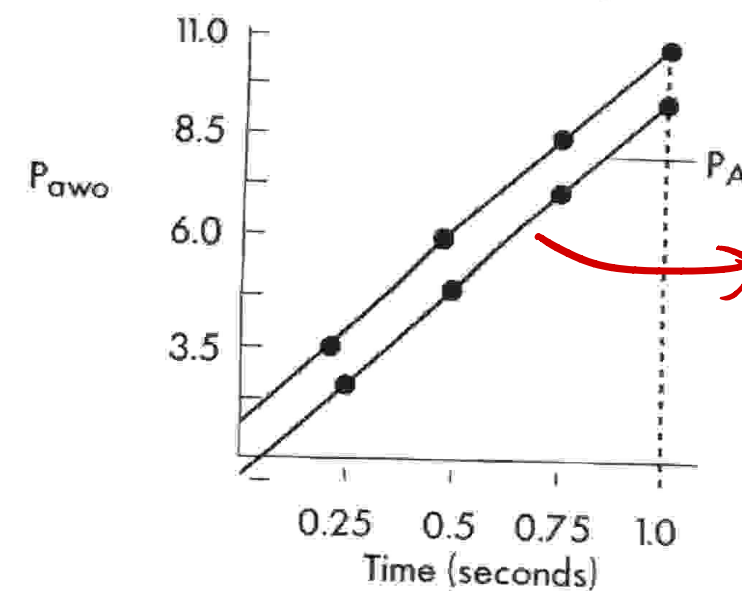
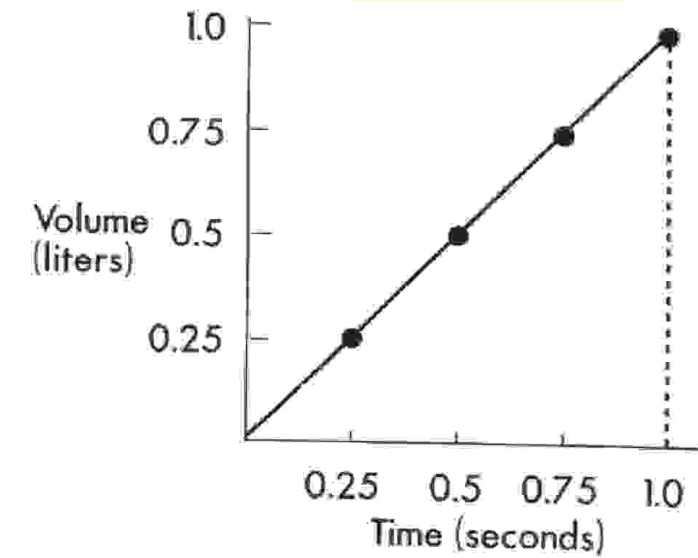
PTA 先出來
Pawo 再出來

若打的 flow stable
PTA 則 stable

$$P_{TA} = \text{flow} \times R_{aw}$$



60 L/min 打方型波



$\rightarrow P_{alv}$ (肺泡壓沒有放機器看不到)
虛指線

Figure 4-2. A high pressure source ventilator with high internal resistance. Flow is constant. Volume rises as an ascending ramp during inspiration. Alveolar pressure (P_A) also rises as an ascending ramp. Transairway pressure (P_{TA}) is a rectangular wave. Monitored gauge pressure at the airway opening (P_{awo}) is the sum of P_{TA} and P_A . See Box 4-3 for an example.

Plateau Pressure ($P_{plateau}$)

肺部順應性 { ①靜態
②動態



Definition

→ 沒有flow的時候
A **static** pressure measurement taken after a breath has been delivered, but before exhalation begins.

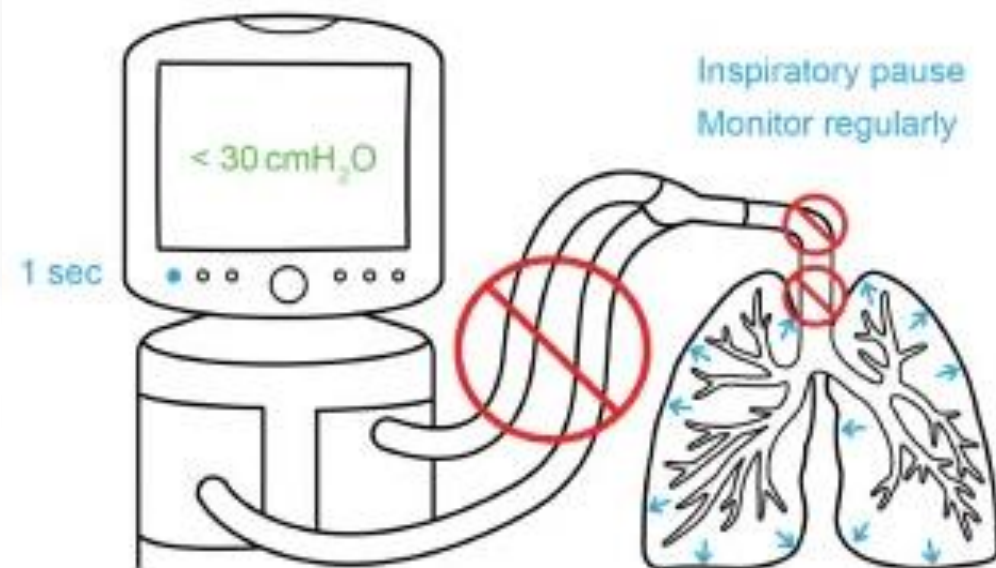
Physiological State

Reflects the effect of elastic recoil on the gas volume strictly inside the alveoli (Intrapulmonary Pressure) under strictly ~~no~~ **no-flow** conditions.

Executing the Plateau Pressure Measurement

Mechanism

Activate an **inflation hold** or **inspiratory pause** on the mechanical ventilator interface.



Duration

Exhalation is mechanically prevented for a brief moment.

0.5 to 1.5 seconds

Requirement

The patient must not be actively breathing. Active respiratory muscle effort renders this static compliance measurement completely inaccurate.

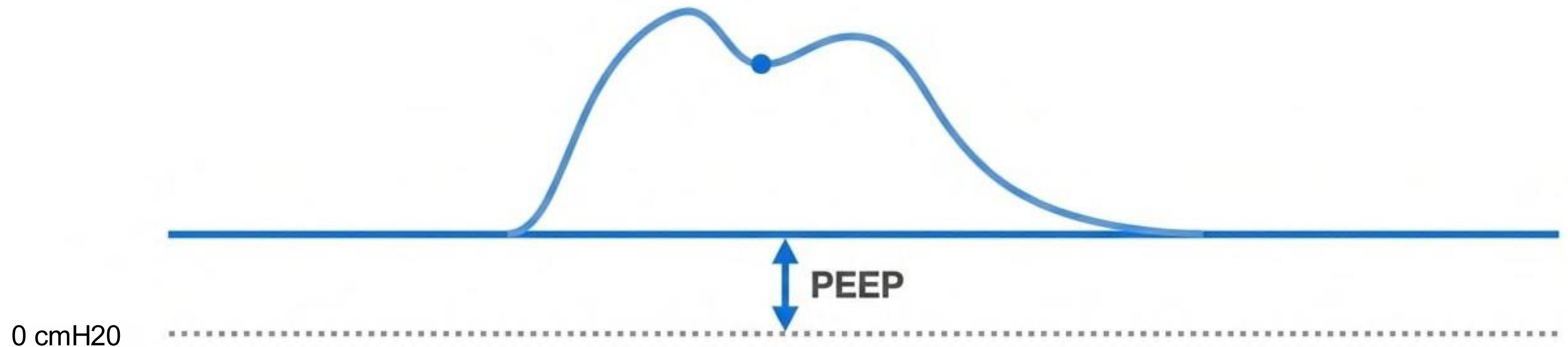
Safety Limits & VILI Prevention

量靜態肺順應性

$P_{plat} < 28 \text{ cmH}_2\text{O}$

Maintaining this strict limit minimizes the development of Ventilator-Induced Lung Injury (VILI). High plateau pressures indicate dangerous alveolar overdistension.

Baseline Pressure and PEEP



Baseline Concept

The pressure at the end of exhalation. While naturally zero (atmospheric) in human physiology, in PPV it is intentionally elevated.

PEEP Definition

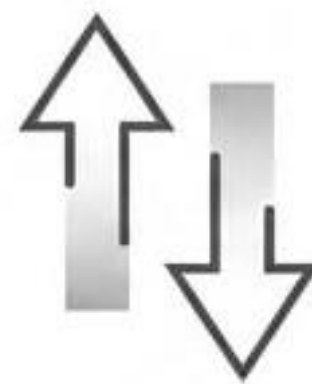
Positive End-Expiratory Pressure. Mechanically applied pressure that prevents the patient from exhaling entirely back to zero.

Physiological Benefits of PEEP



Volume Retention

Increases Functional Residual Capacity (FRC) by maintaining a greater volume of gas in the lungs at end-exhalation.



Pressure Gradient

Maintains a positive end-expiratory transpulmonary pressure.

$$P_L = P_{alve} - P_{pl}$$

肺泡 Open



Alveolar Stability

Prevents the collapse of alveoli (atelectasis) and substantially improves baseline oxygenation.

Extrinsic PEEP

(Therapeutic / Intended)

Mechanically applied baseline pressure set by the clinician to stabilize alveoli.

Intrinsic PEEP (Auto-PEEP)

(Pathological / Accidental)

Definition: Unintentional, accidental air trapping in the lung architecture.

Mechanism: Occurs when a patient does not have adequate expiratory time (T_e) to exhale completely before the ventilator delivers the next mandatory breath.

Risk Profile: Leads to dynamic hyperinflation and negative hemodynamic cardiovascular effects.

若病人沒有足夠的吐氣時間會有此狀況

Auto-PEEP (Intrinsic PEEP)

有Auto PEEP 吸氣會更費力

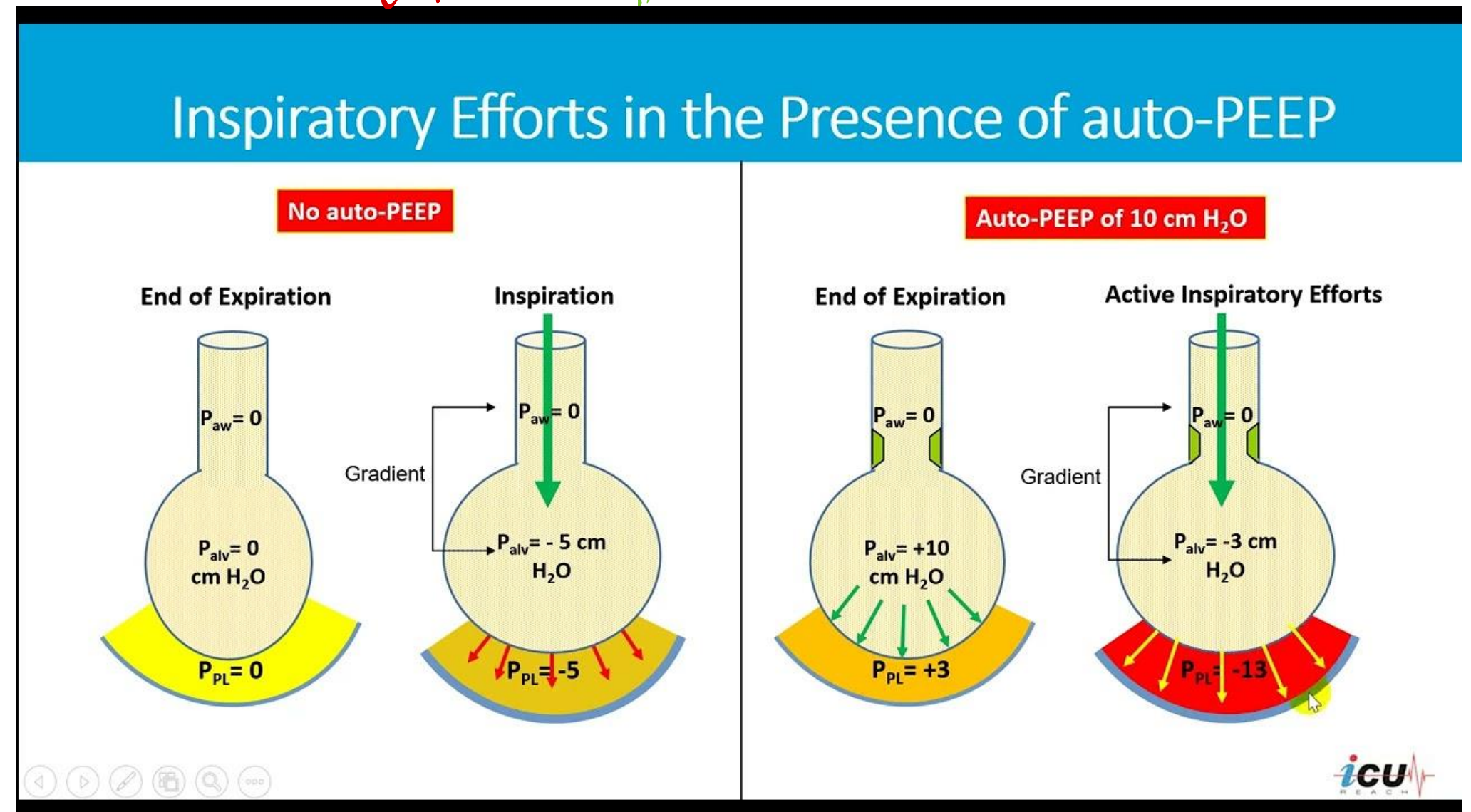
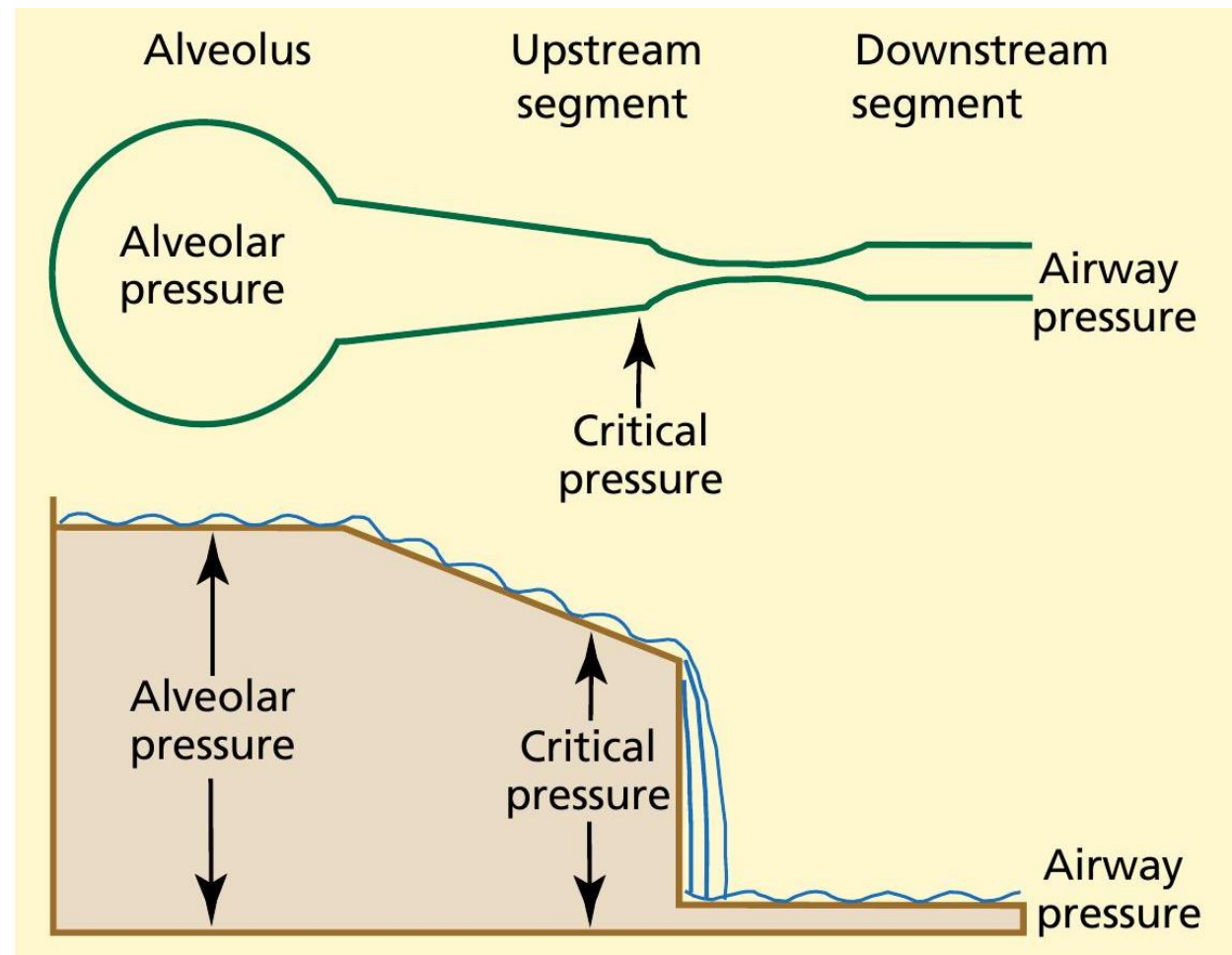
小氣道過早關閉

因為氣道分裂、發炎、腫

• 氣喘

• COPD

處制: broncal dilater, steroid



Division of the Inspiratory and Expiratory Periods

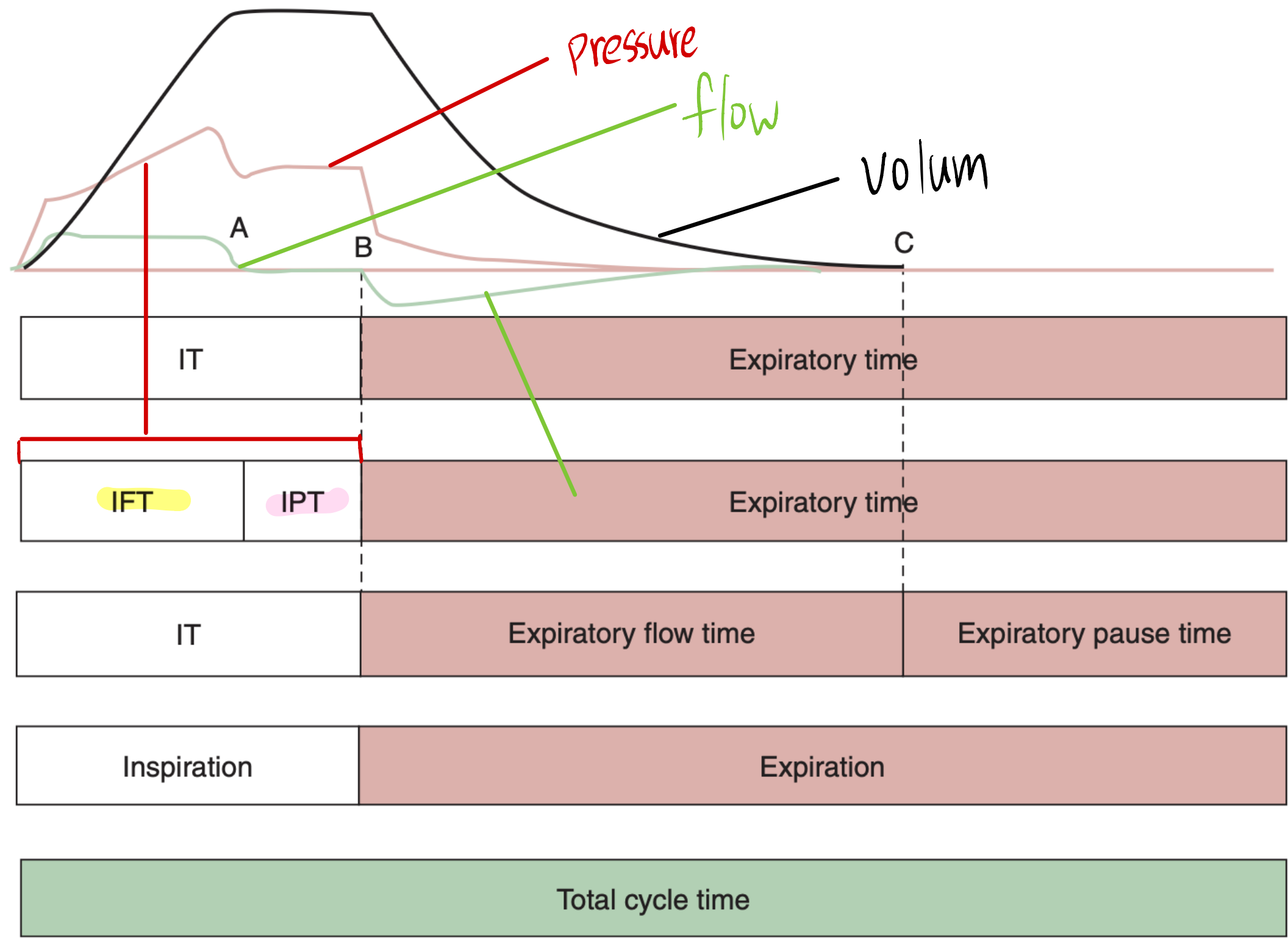
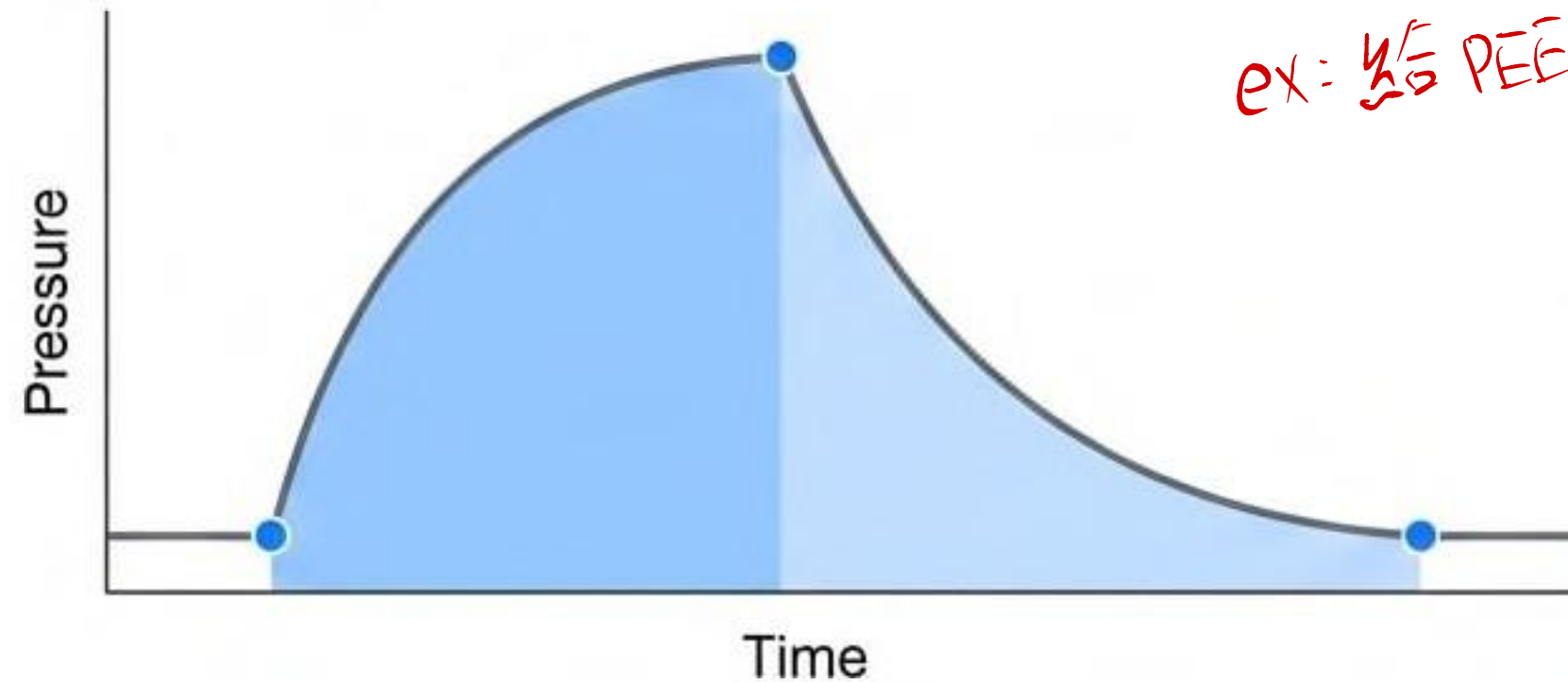


FIGURE 3-20 Divisions of the inspiratory and expiratory periods. A volume-controlled breath is depicted. **A.** End of inspiratory flow. **B.** Start of expiratory flow. **C.** End of expiratory flow. **IFT**, Inspiratory flow time; **IPT**, inspiratory pause time; **IT**, inspiratory time.

Mean Airway Pressure (Paw / MAP)

增加 MAP 可以改善氧合
ex: 给 PEEP



Definition

The average pressure recorded in the airway throughout the entire respiratory cycle—combining both inspiratory and expiratory phases.

Clinical Link

MAP is the primary determinant of **Oxygenation** during positive pressure ventilation.

$$\text{MAP} = [\text{Mean inspiratory Paw} \times (\text{Ti} / \text{TCT})] + [\text{PEEP} \times (\text{Te} / \text{TCT})]$$

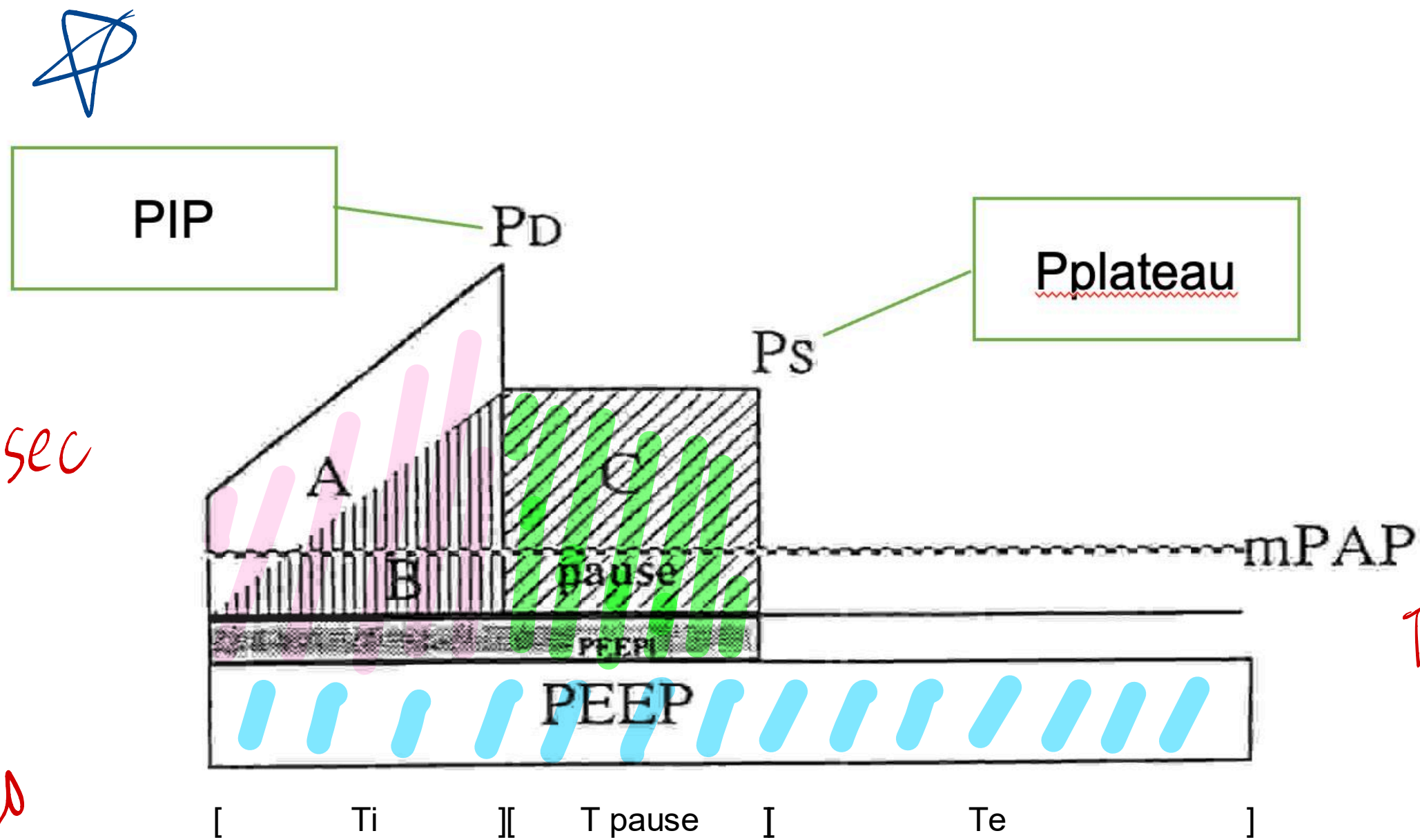
Rate : 15 bpm

$$T_i = 1 \text{ sec}$$

$T_i - P_{\text{plateau}} = 0.5 \text{ sec}$

$$PIP = 0.5$$
$$P_{pla} = 20$$

PEEP = 5

$$\tau_e = 2.5 \text{ sec}$$


$$I + E = 4$$

4-1.5

$$T_e = 2.5$$

Inspiratory Time = $T_i + T_{\text{pause}}$

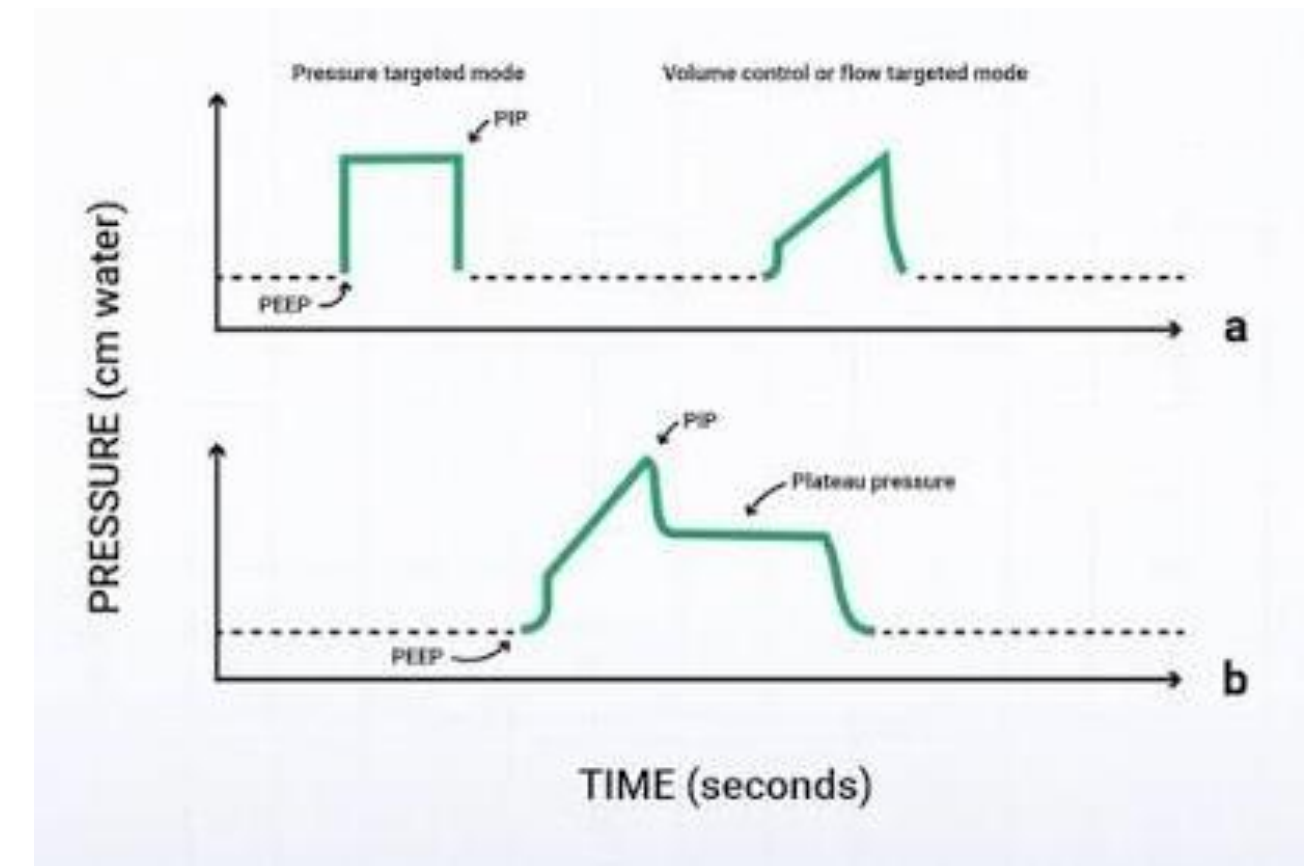
$$I : E = (T_i + T_{\text{pause}}) : T_e$$

$$\text{MAP} = \frac{1}{2} [\text{PIP} * \text{Ti/TCT}] + [\text{Pplat} * \text{T pause/TCT}] + [\text{PEEP} * \text{Te/TCT}]$$

$$\text{MAP} = \frac{1}{2} [\text{PIP} - \text{PEEP}] * \text{Ti/TCT} + [\text{Pplat} - \text{PEEP}] * \text{T pause/TCT} + \text{PEEP}$$

Factors That Increase Mean Airway Pressure

- Increasing level of PEEP
- Increasing inspiratory pressure
- Increasing duration of inspiration (T_i)
- Nature of the inspiratory waveform
- Decreasing compliance, increasing resistance



The Equation of Motion of the Respiratory System

$$\underbrace{P_{vent}}_{\text{呼吸器}} + \underbrace{P_{mus}}_{\text{呼吸肌壓力}} = \underbrace{(V / C)}_{\text{肺泡壓}} + \underbrace{(R * \dot{V})}_{\substack{\text{flow} \\ P_{TA}}}$$

The Foundational Relationship

Defines the interaction between ventilator pressure (P_{vent}), muscular effort (P_{mus}), compliance (C), resistance (R), flow ($\dot{V}$), and volume (V).

Elastic Load (V / C)

The pressure required to overcome the elastic recoil of the lungs and chest wall.

Resistive Load ($R * \dot{V}$)

The pressure required to overcome the frictional resistance of the airways

Volume Delivery under Constant Flow

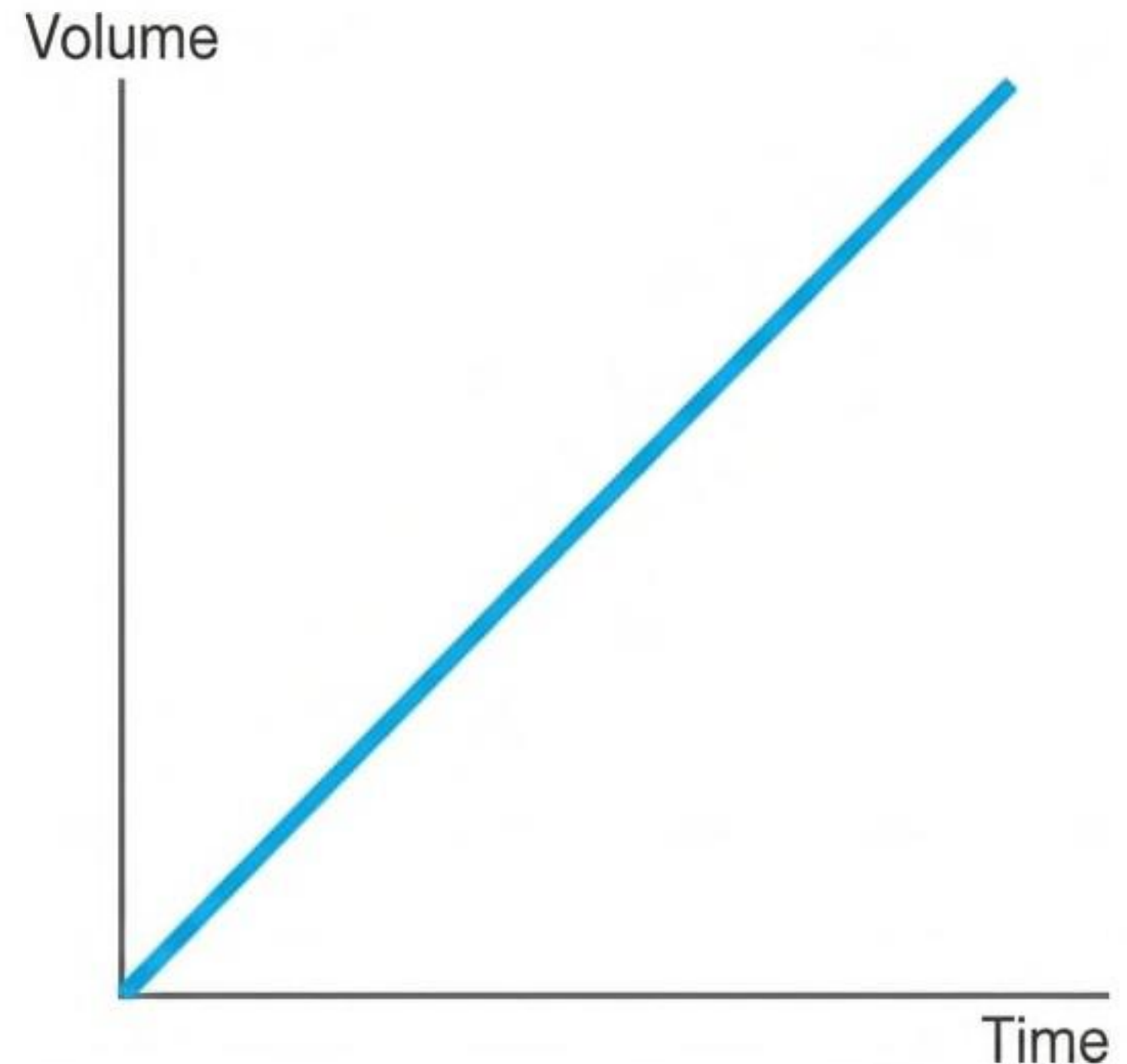
$$V = \text{Flow} * T_i$$

Linear Accumulation

Volume (V) delivered is a direct, linear function of the preset flow and the given inspiratory time (T_i).

Steady State Delivery

In constant flow ventilation, the delivered volume increases at a steady, uninterrupted rate throughout the entire inspiratory phase.



Setting and Calculating Inspiratory Time (Ti)

$$T_i = V / \text{Flow}$$

平均吸氣流量

Deriving the Time Variable

Inspiratory time is precisely calculated by dividing the desired tidal volume by the set flow rate.

Crucial Unit Conversion

Flow must be converted from L/min to L/s for accurate calculation (e.g., a flow of 60 L/min equals exactly 1 L/s).

Clinical Example

Delivering a tidal volume of **0.5 L** at a flow of **60 L/min (1 L/s)** takes exactly **0.5 seconds**.

Gas Flow Dynamics & Pressure Gradients

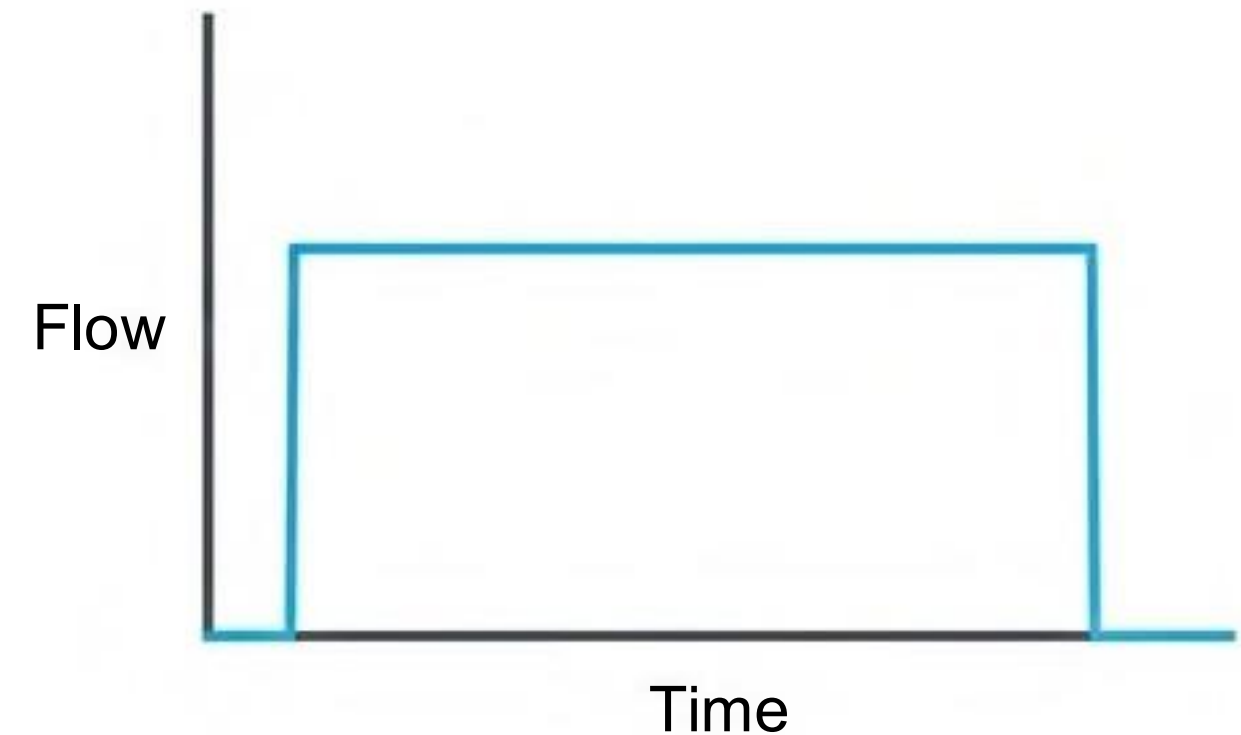
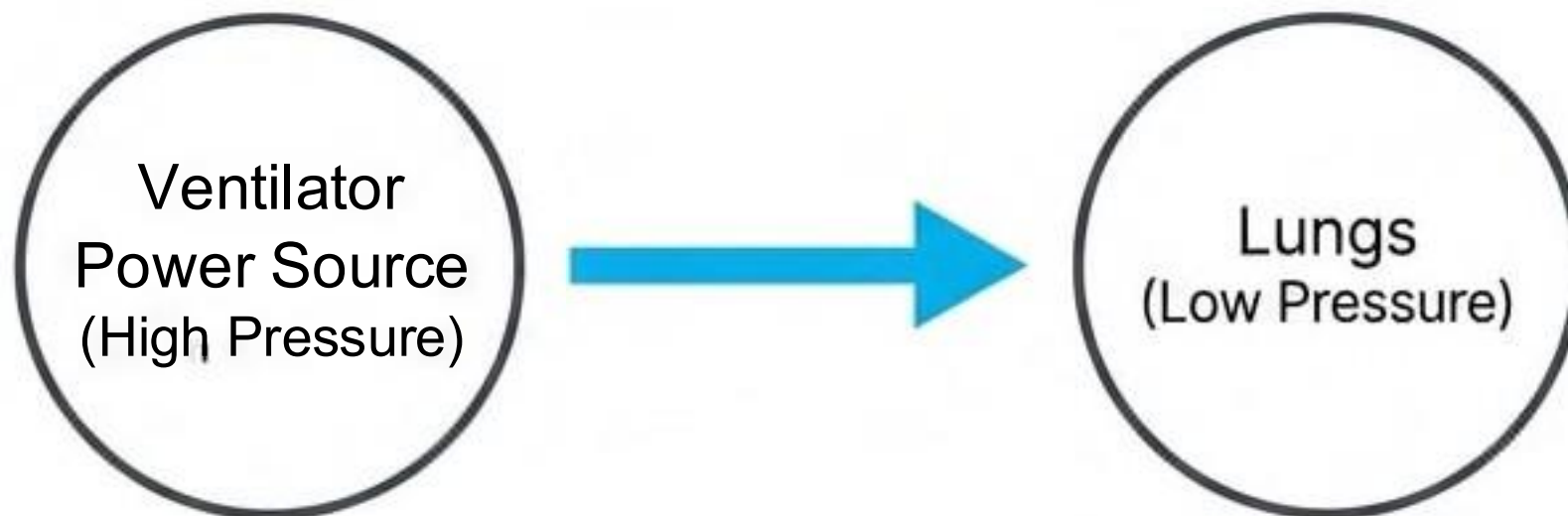
$$\text{Flow} = V / T_i \quad (\text{L/min or L/sec})$$

The Driving Force of Flow

Gas flow depends directly on the pressure difference between the ventilator power source and the pressure inside the lungs.

Pressure Gradient Proportionality

The larger the pressure gradient, the faster the flow of gas and the sooner the lungs reach full tidal volume.



Alveolar Pressure Dynamics under Constant Flow

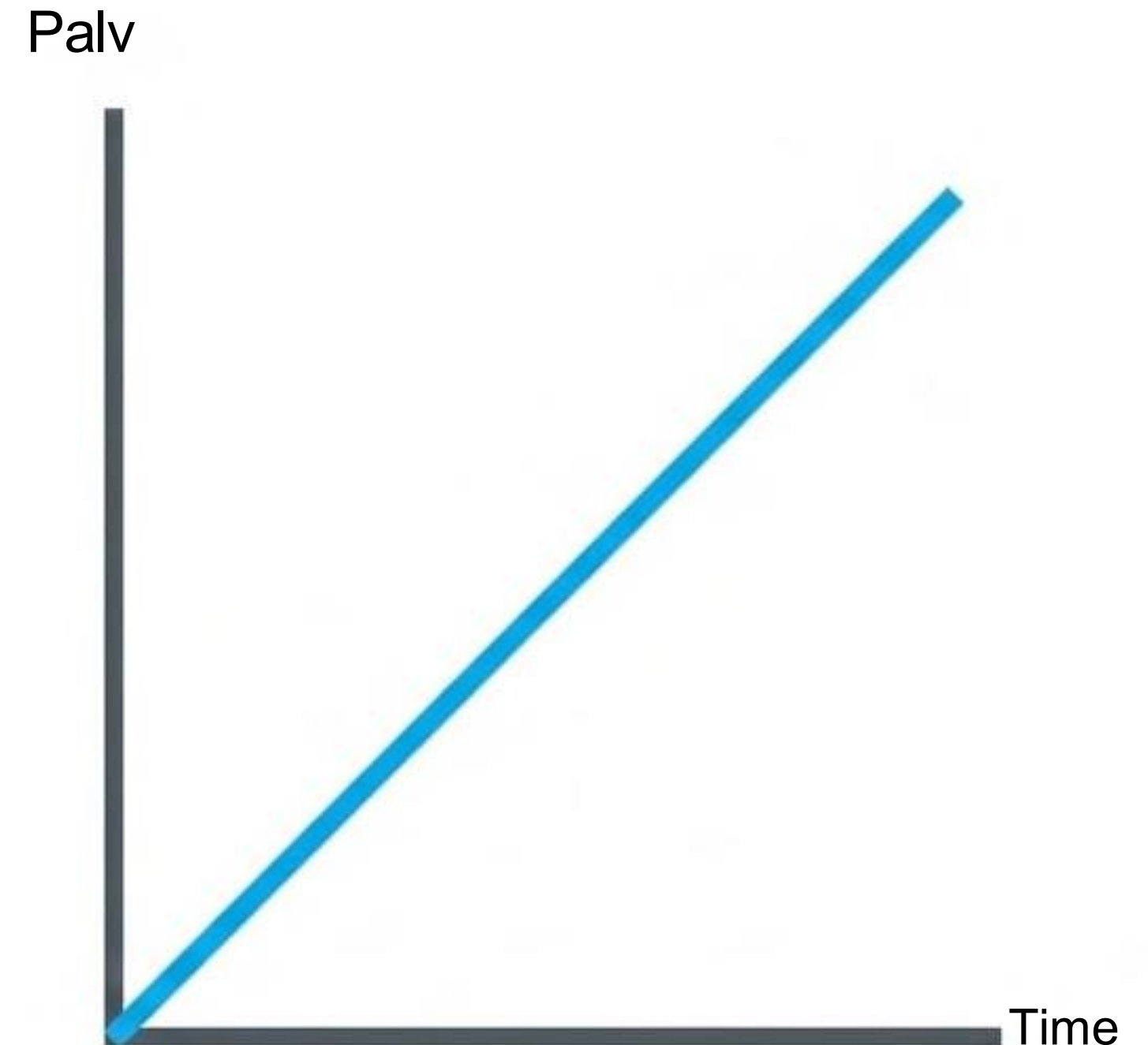
$$P_{alv} = V/C$$

Volume Accumulation vs. Stiffness

Alveolar pressure (P_{alv}) is purely dependent on lung stiffness (Compliance, C) and the accumulated gas volume.

The Linear Ramp

Because volume increases linearly under constant flow, P_{alv} rises as a steady, linear ramp, reaching its maximum peak exactly at end-inspiration.



Transairway Pressure Behavior under **Constant Flow**

$$\text{PTA} = \text{Flow} * R_{aw}$$

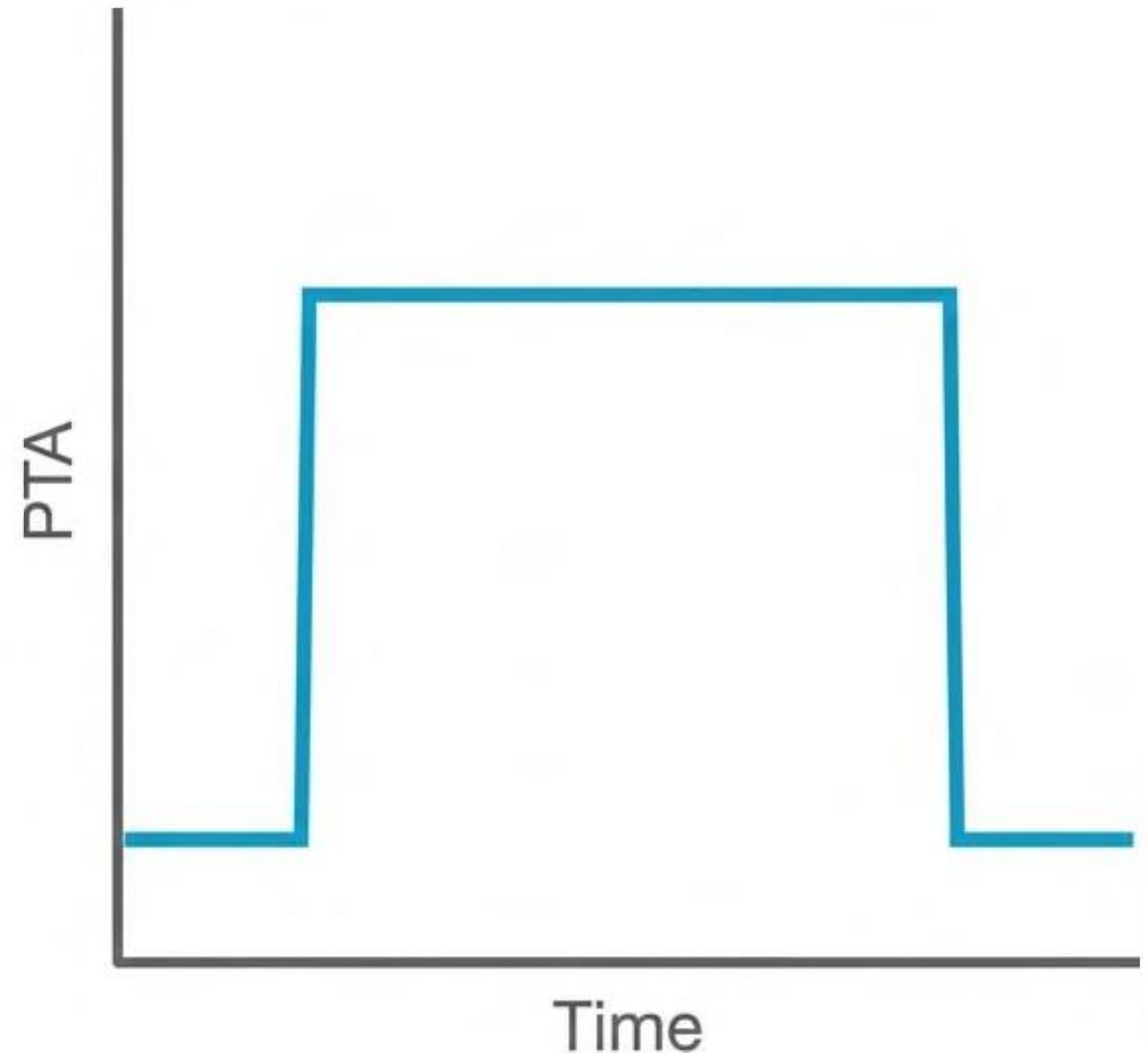
(or PIP - Pplateau)

Isolation of Frictional Forces

PTA represents the pressure lost strictly to airway resistance (R_{aw}), entirely separate from lung compliance.

Complete Constancy

Because the flow is mechanically held constant, the PTA remains completely flat and constant throughout the entire inspiratory phase.



The Airway Pressure Curve (Paw)

$$P_{aw} = P_{TA} + P_A$$

The Summation of Forces

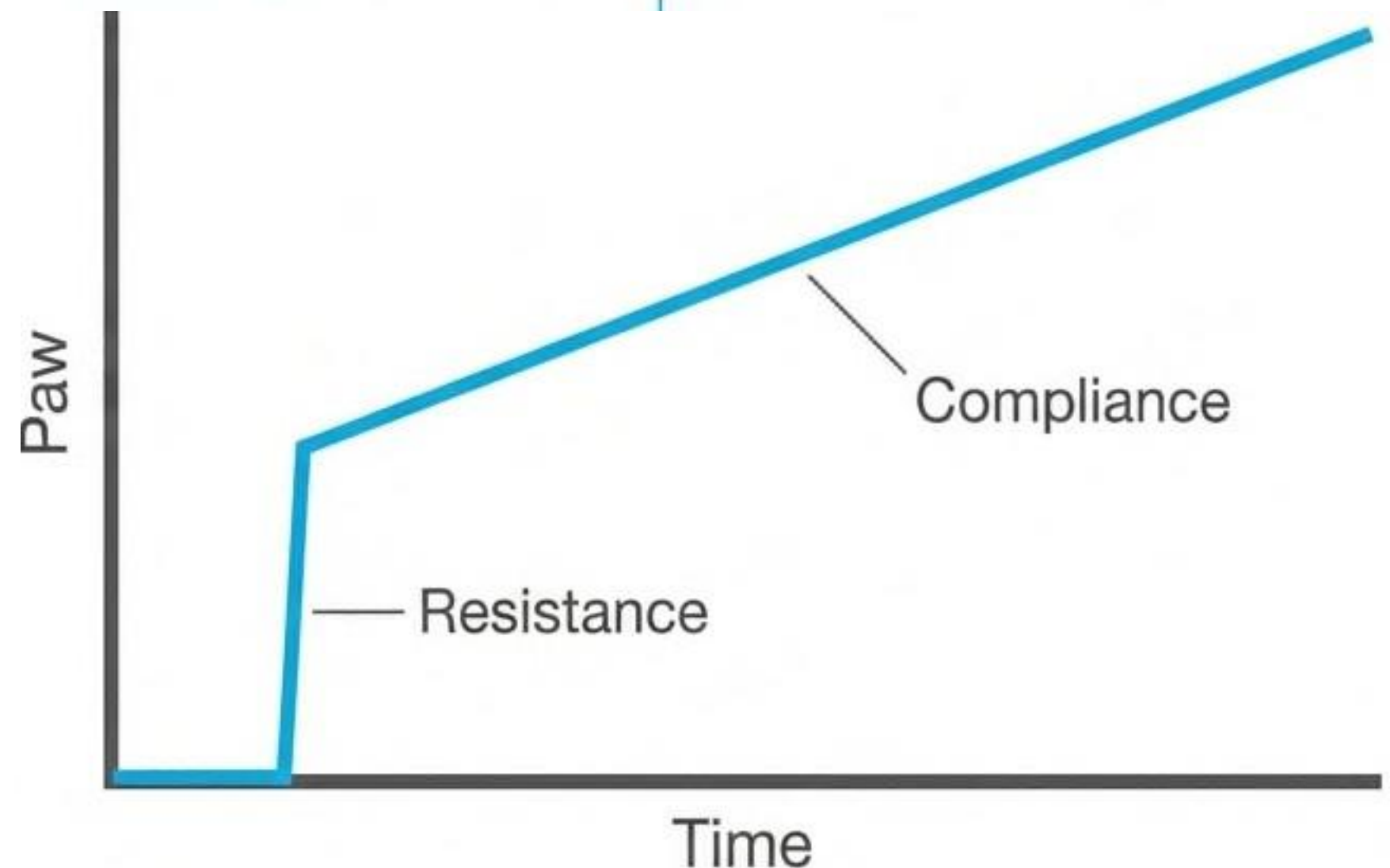
Total airway opening pressure (Paw) is the literal summation of the resistive component and the elastic component.

Step Height (Resistance)

The initial vertical step in the curve is determined by PTA (driven by Flow and Raw).

Climbing Slope (Compliance)

The subsequent climbing slope is determined by Palv (driven by Volume and Compliance).



Opposing Forces in Lung Inflation

The total opposition to lung inflation is governed by two primary physical forces.

彈性

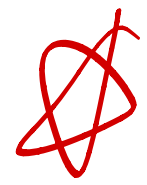
Elastic Forces

Arise from the inherent elastic properties of the lungs and the chest wall.

阻力

Frictional Forces

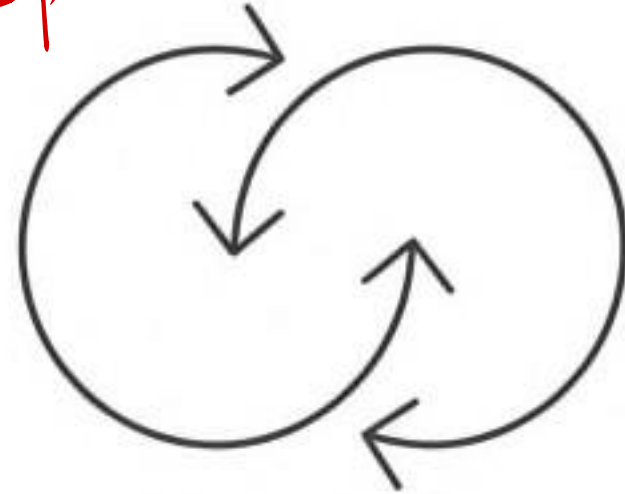
Composed of tissue viscous resistance and airway resistance to gas flow.



Compliance and Elastance

Compliance (C) = $\Delta \text{Volume} / \Delta \text{pressure}$

Represents the relative ease with which a structure distends.



Elastance (e) = $\Delta \text{pressure} / \Delta \text{Volume}$

The physical tendency of a structure to return to its original form after being stretched.

$$\text{Compliance} = \frac{\Delta V}{\Delta P} = \frac{V_T}{\text{Driving pressure}} = \frac{V_T}{P_{\text{pla}} - \text{PEEP}}$$

$$C = 1 / e$$

$$e = 1 / C$$

Clinical Compliance Formula

$$C = \frac{\Delta V}{\Delta P}$$

Compliance of the respiratory system measures the volume change per unit of applied pressure. It represents the combined elastic compliance of the lung parenchyma and the surrounding thoracic structures.

ex: $P_{pla} = 20$ $PEEP = 5$

$V = 500 \text{ ml}$

$C = 500 / 15 = 33.33$

Normal Compliance Ranges

Spontaneously Breathing

50 to 170 mL/cmH₂O

Average: 100 mL/cmH₂O

Intubated Males

40 to 50 mL/cmH₂O

Up to 100 mL/cmH₂O

Intubated Females

35 to 45 mL/cmH₂O

Up to 100 mL/cmH₂O

Static Compliance (Cs) 取肺泡

靜態

$$Cs = VT / (P_{plateau} - \underline{EEP})$$

吐氣末壓 < PEEP 有/無
or
Auto PEEP

- Calculated under pure static (no-flow) conditions.
- Requires an end-inspiratory pause maneuver to isolate pure alveolar elastic recoil.
- The patient must be completely passive for accurate measurement.

Dynamic Compliance (Cd) 動態 取肺部

$$Cd = VT / (PIP - EEP)$$

Measurement Conditions

- Calculated during active gas flow.
- Reflects both the elastic recoil of the respiratory system and the dynamic resistive load of the airways.

Clinical Rule: $Cd < Cs$

Pathological Compliance Scenarios

Decreased Compliance	Increased Compliance
<i>Stiff Lungs</i>	<i>Flabby Lungs</i>
Requires higher pressure to achieve volume. Conditions: ARDS, pneumonia, pulmonary fibrosis, or <u>kyphoscoliosis</u> .	Inflates easily but recoils poorly due to loss of elastic tissue. Conditions: Pulmonary emphysema.

胸壁限制的關係

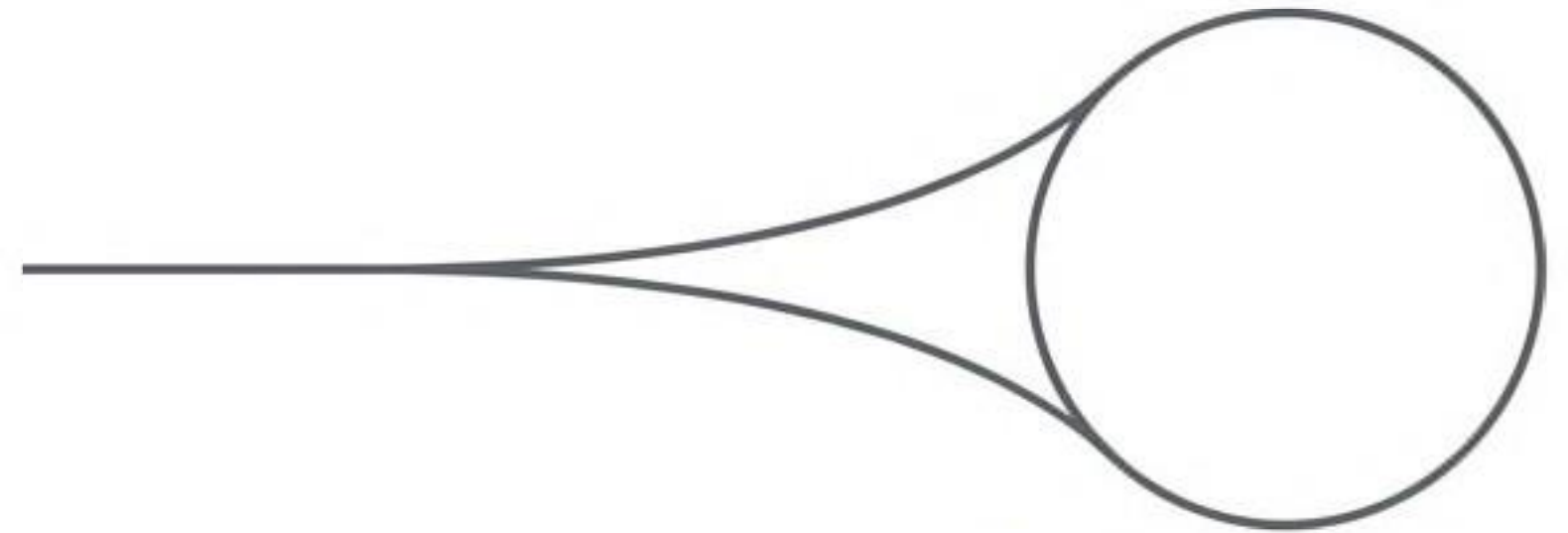
Airway Resistance (R_{aw})

Resistance is the frictional opposition to gas flow through conductive airways.

Governed by Poiseuille's law.

Depends on:

1. Gas viscosity
2. Gas density
3. Tube length
4. **Airway radius**



Clinical Rule:

Frictional resistance is **zero**

when flow is zero.

Airway Resistance Calculation

$$\text{Raw} = (\text{PIP} - \text{Pplateau}) / \text{Flow}$$

or Raw = PTA / Flow

- Quantified using peak inspiratory pressure (PIP) and plateau pressure (Pplateau).
- Expressed in cmH20/(L/s).
- Crucial: Flow must be converted from L/min to L/s before calculation.

Normal Airway Resistance Values

Unintubated

0.6 to 2.4
cmH₂O/(L/s)

Measured at 0.5 L/S flow

Intubated

5 to 7
cmH₂O/(L/s)

Increases as ET tube size decreases

Obstructive Diseases

13 to 18+
cmH₂O/(L/s)

Asthma & COPD patients

The Time Constants

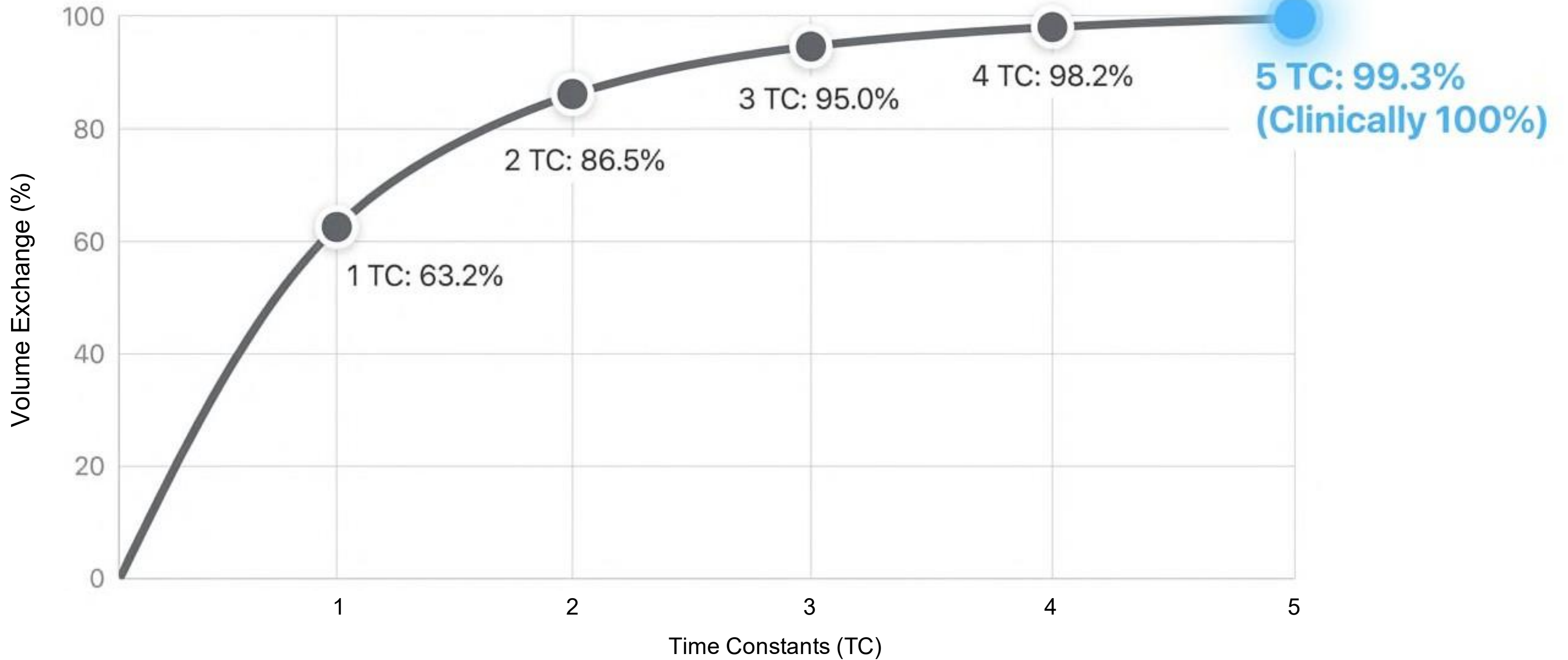
Time Constants (s) = Compliance * Resistance

$$\Delta V / \Delta P$$

$$L/s$$

The Exponential Law of Passive Exchange

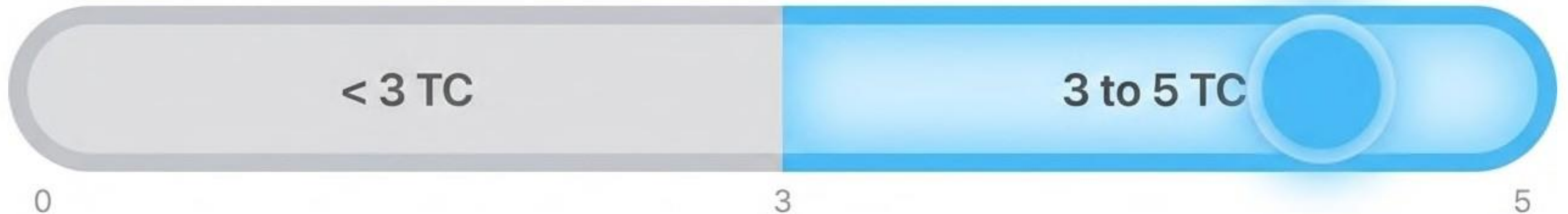
1 Time Constant = Compliance x Resistance



Defining the **Inspiratory Window (Ti)**

The exact inspiratory time depends entirely on the patient's specific time constants.

Threshold Dashboard



Below Threshold

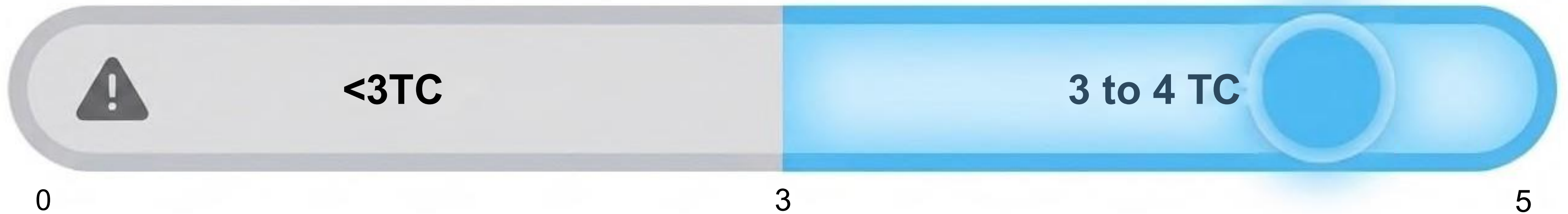
Results in incomplete tidal volume delivery and inadequate ventilation.

Optimal Clinical Target

Ensures even gas distribution across heterogeneous units and guarantees complete, adequate volumetric delivery.

Defining the **Expiratory Threshold (Te)**

Threshold Dashboard



Below Threshold

Leads to incomplete lung emptying, an artificial increase in Functional Residual Capacity (FRC), and dangerous air trapping (auto-PEEP).

Optimal Target

Allows 95% to 98% of the volume to empty, which is clinically adequate and safe for passive exhalation.

The Respiratory System is Not Homogeneous



Core Principle

The lung is a complex, uneven landscape. Regional compliance and resistance vary significantly between acinar units.

The Consequence

Gas distribution, as well as the exact rate of filling and emptying, depend entirely on these localized physical properties.

Fast vs. Slow Lung Units

Fast Units

Profile:	Short Time Constant
Mechanics:	Decreased compliance, normal/low resistance
Behavior:	Fill and empty rapidly
Requirement:	Demands high pressure to achieve volume
Example:	Pulmonary Fibrosis

Slow Units

Profile:	Long Time Constant
Mechanics:	Increased resistance and/or compliance
Behavior:	Fill and empty slowly
Requirement:	Highly prone to air trapping if T_e is too short
Example:	Emphysema

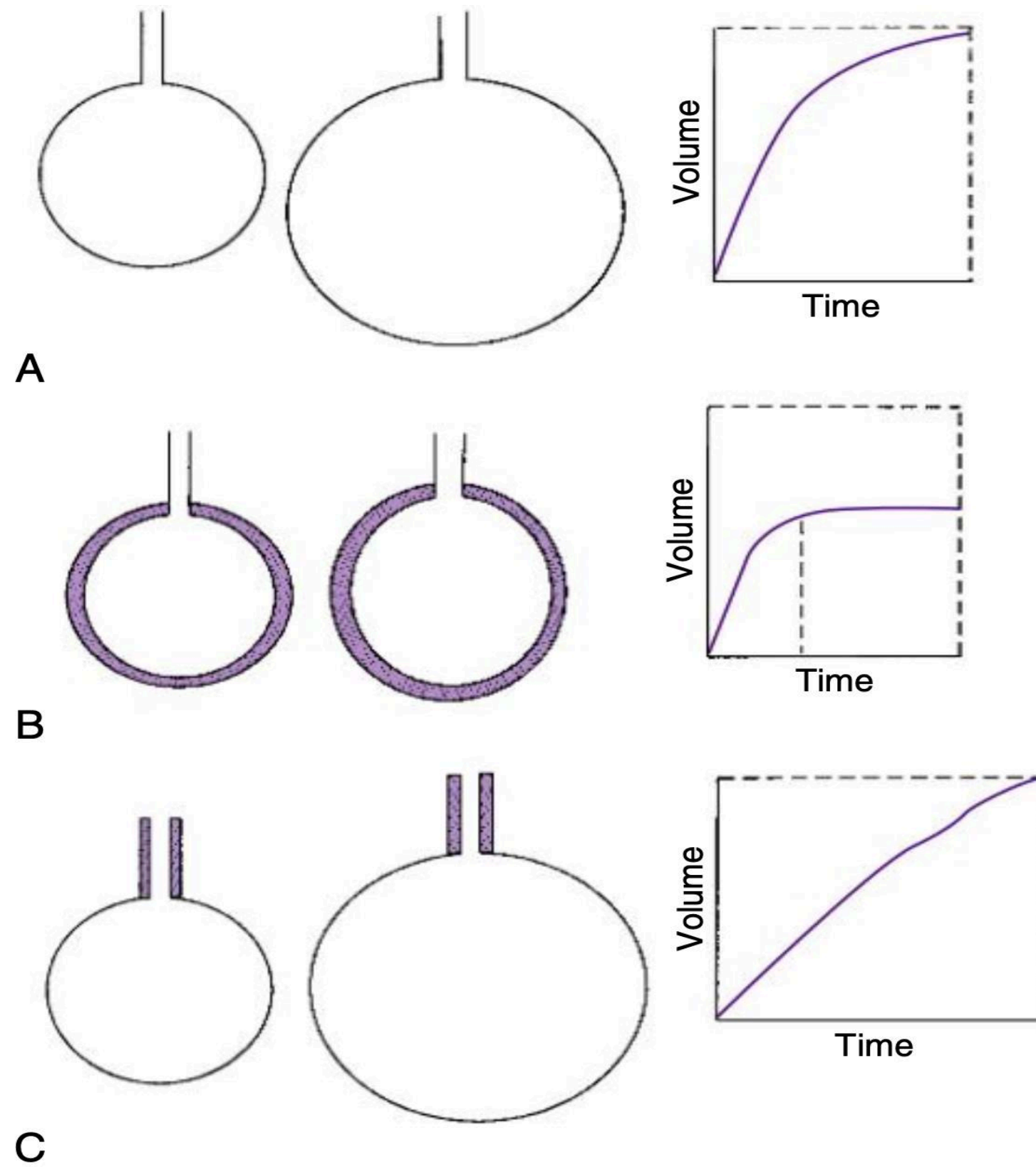
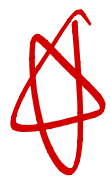


Fig. 1.5 (A) Filling of a normal lung unit. (B) A low-compliance unit, which fills quickly but with less air. (C) Increased resistance; the unit fills slowly. If inspiration were to end at the same time as in (A), the volume in (C) would be lower.



Effects of PPV on Ventilation: Alveolar Ventilation & PaCO₂

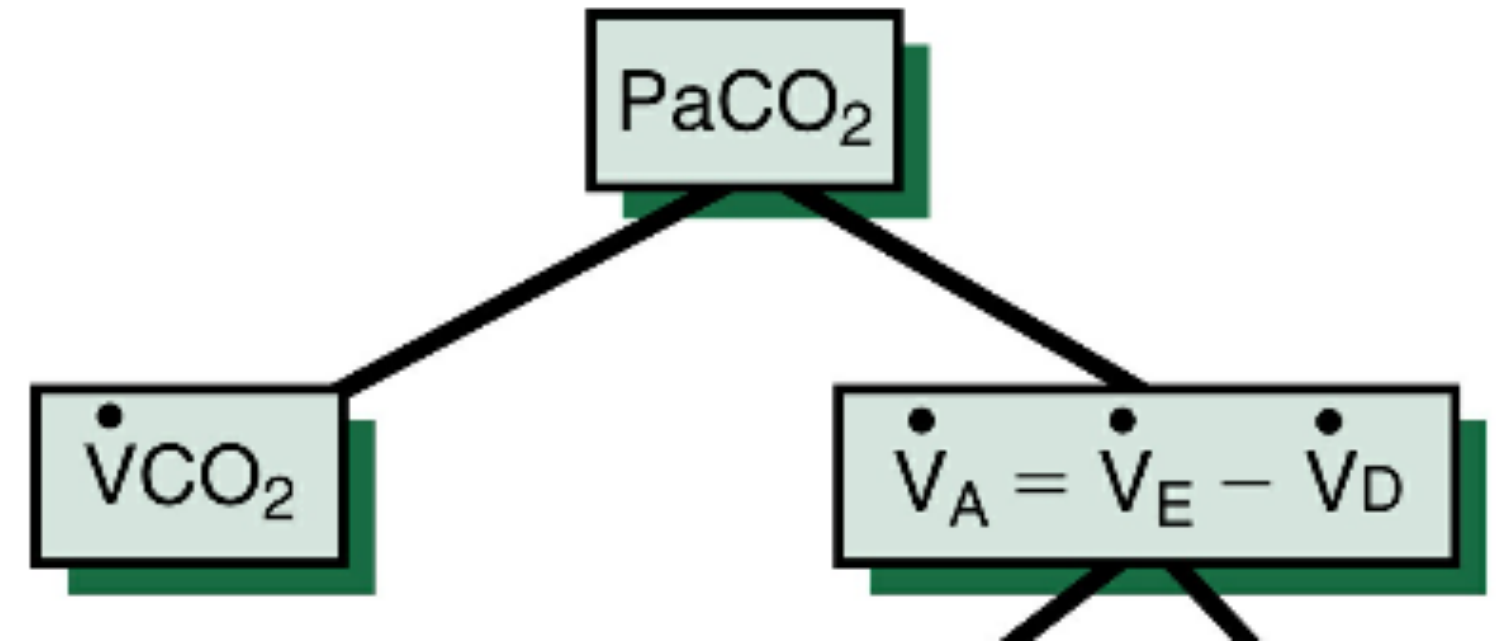
Increased minute ventilation (VE)

$$- VE = f * VT$$

Increased alveolar ventilation (VA)

$$- VA = f * (VT - VD)$$

理想體重 → 磅數 = Dead space



$$PaCO_2 = (VCO_2 * 0.863) / VA$$

$$VA = (VCO_2 * 0.863) / PaCO_2$$

Anatomic Dead Space

Baseline anatomical non-participating volume.

**Normal Value:
1 mL/lb IBW or
2.2 mL/kg IBW**

Physiologic Dead Space

Positive Pressure Ventilation (PPV) artificially increases physiologic dead space via:

- Capillary compression.
- Alveolar hyperinflation.
- Decreased pulmonary perfusion.

Mechanical Volume Loss

Compressible volume lost in circuit tubing reduces the effective tidal volume delivered to the patient.

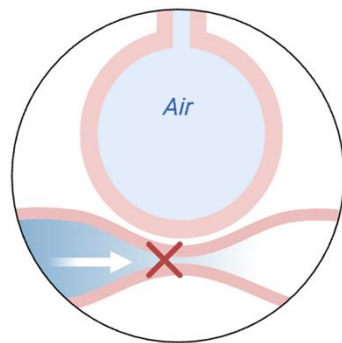
 **Volume Lost =
Tubing Compliance x
(PIP - PEEP)**

Impact on Ventilation-Perfusion (V/Q) Matching

$$V/Q > 1$$

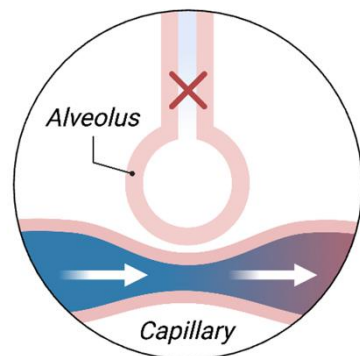
Dead space ventilation

Ventilation
without perfusion

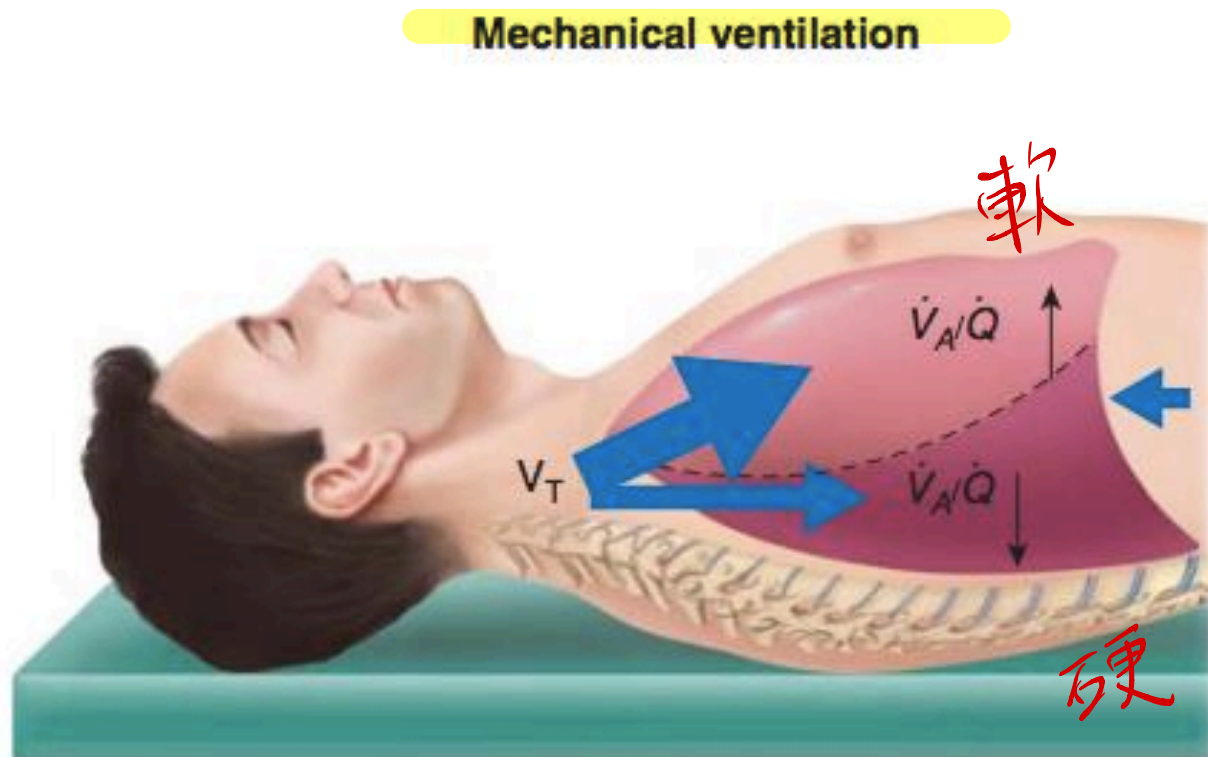


Shunt perfusion

Perfusion without
ventilation

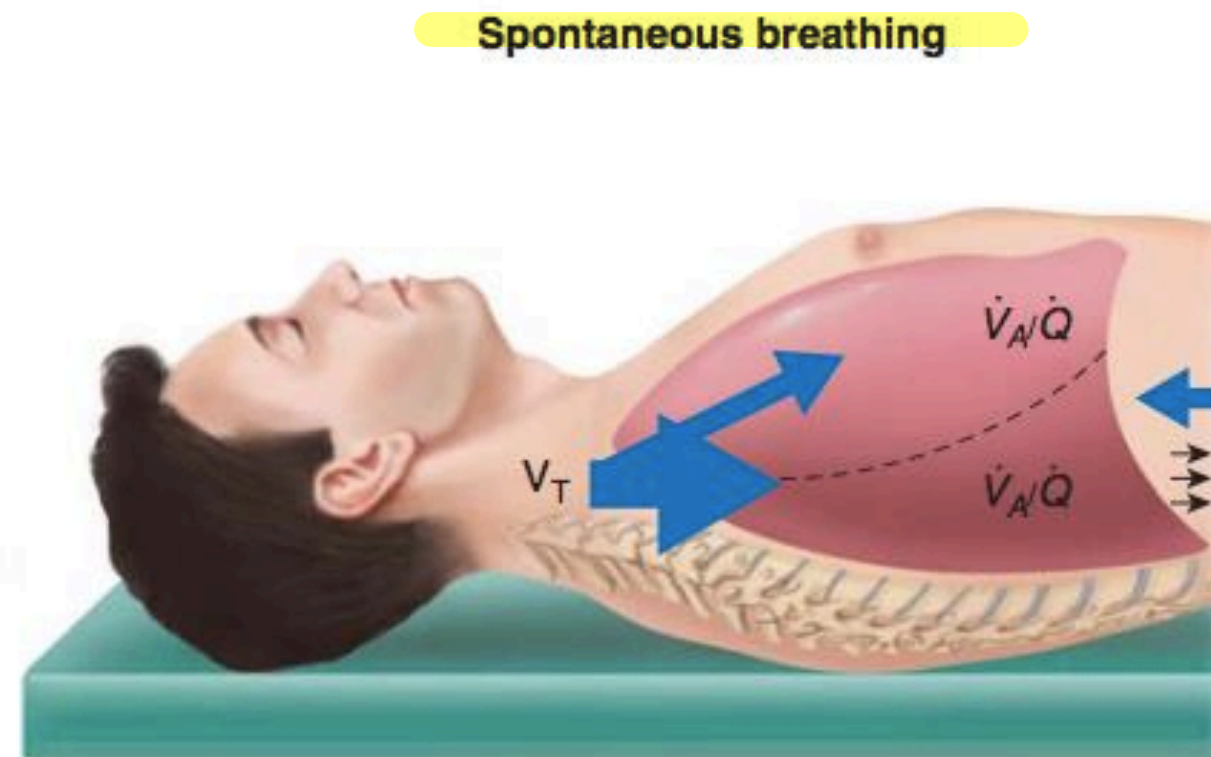


$$V/Q < 1$$



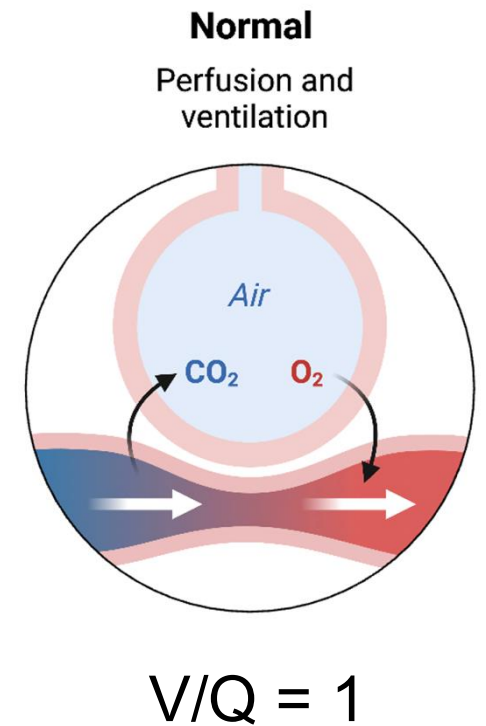
Result: **V/Q decreased.**

V/Q mismatch



Result: Normal and highly efficient V/Q matching.

→ 氣合不好



Effects of PPV on Oxygenation

FiO₂

Fraction of Inspired Oxygen

Directly raises alveolar oxygen tension (P_AO_2), thereby increasing the driving pressure for oxygen diffusion across the alveolar-capillary membrane.

MAP

Mean Airway Pressure

The primary determinant of oxygenation during PPV. It represents the average pressure applied to the airways throughout the entire respiratory cycle.

PEEP

Positive End-Expiratory Pressure

Maintains positive end-expiratory transpulmonary pressure to:

- Increase Functional Residual Capacity (FRC).
- Recruit collapsed alveoli.
- Reduce pulmonary shunting.

$V/Q < 1$

Clinical Mathematics in Practice

Inputs: VC Mode

$V_T = 750 \text{ mL}$

$f = 15 \text{ bpm}$

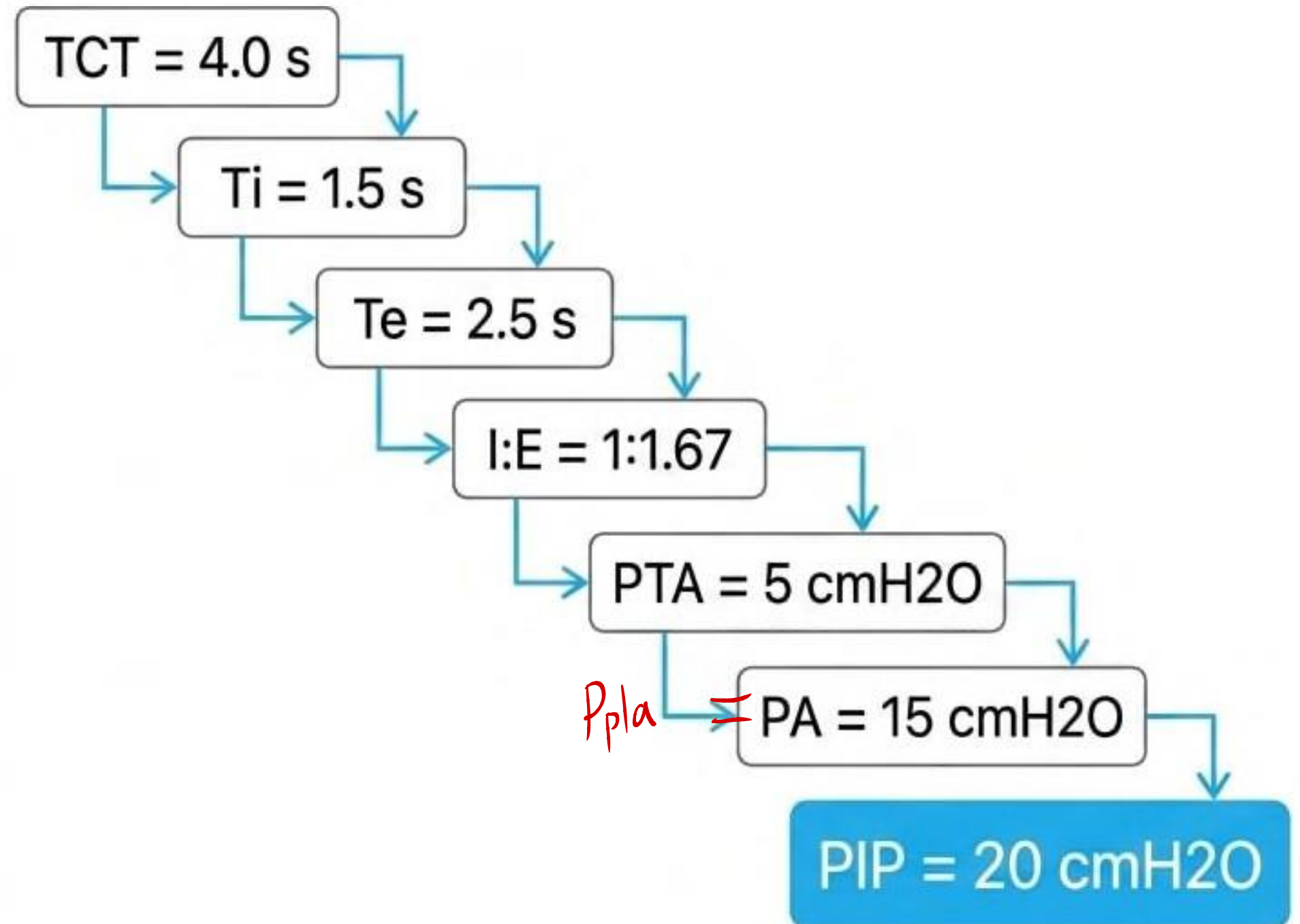
Flow = 30 L/min (0.5 L/s)

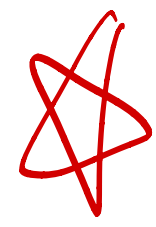
Raw = 10 cmH₂O/L/s

$C_s = 50 \text{ mL/cmH}_2\text{O}$

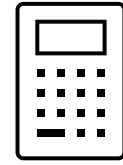
$$PIP = PTA + P_{pla}$$

Calculated Outputs





Derivative Calculation



What would be the impact if **resistance doubled**?

If **compliance** were **reduced to 50%** of its original value, how would PA, PTA and PIP be affected?

$$PA = \frac{750}{25} = 30$$

$$PIP = 30 + 5 = 35$$

$$\begin{aligned} R_{aw} &= 10 \rightarrow 20 \text{ cmH}_2\text{O} \\ PTA &= 20 \times 0.5 = 10 \text{ cmH}_2\text{O} \\ PIP &= 10 + P_{pla} = 10 + 15 = 20 \end{aligned}$$

$$C = \frac{\Delta V}{\Delta P}$$

$$P_{pla} = \Delta P + PEEP$$

V/Q mismatch $\Rightarrow$ most common cause of hypoxemia

Normal V/Q matching = 1

V: ventilation

Q: perfusion

1. Low V/Q, $V/Q < 1$

① $V \downarrow$ $V/Q \Rightarrow Q/Q \Rightarrow$ cause shunt

- Asthma
- COPD
- Pneumonia
- Atelectasis
- Pulmonary edema
- Mucus plugging
- Aspiration
- ARDS

2. high V/Q, $V/Q > 1$

① $V \uparrow, Q \downarrow$

- pulmonary embolism
- Low cardiac output
- Pulmonary vascular obstruction
- Overdistended alveoli from excessive PEEP
- Emphysema with capillary bed destruction

shunt

$$V/Q = 0/Q$$

dead space

$$V/Q = V/0$$